

NEET 2025

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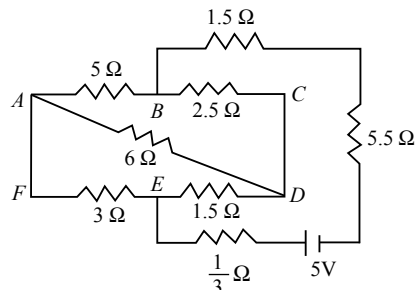
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PHYSICS

- A parallel plate capacitor made of circular plates is being charged such that the surface charge density on its plates is increasing at a constant rate with time. The magnetic field arising due to displacement current is
 - non-zero everywhere with maximum at the imaginary cylindrical surface connecting peripheries of the plates
 - zero between the plates and non-zero outside
 - zero at all places
 - constant between the plates and zero outside the plates.
- An electric dipole with dipole moment $5 \times 10^{-6} \text{ C m}$ is aligned with the direction of a uniform electric field of magnitude $4 \times 10^5 \text{ N/C}$. The dipole is then rotated through an angle of 60° with respect to the electric field. The change in the potential energy of the dipole is
 - 1.2 J
 - 1.5 J
 - 0.8 J
 - 1.0 J
- A ball of mass 0.5 kg is dropped from a height of 40 m. The ball hits the ground and rises to a height of 10 m. The impulse imparted to the ball during its collision with the ground is (Take $g = 9.8 \text{ m/s}^2$)
 - 0
 - 84 Ns
 - 21 Ns
 - 7 Ns
- The intensity of transmitted light when a polaroid sheet, placed between two crossed polaroids at 22.5° from the polarization axis of one of the polaroid, is (I_0 is the intensity of polarised light after passing through the first polaroid)
 - $\frac{I_0}{8}$
 - $\frac{I_0}{16}$
 - $\frac{I_0}{2}$
 - $\frac{I_0}{4}$
- The kinetic energies of two similar cars A and B are 100 J and 225 J respectively. On applying brakes, car A stops after 1000 m and car B stops after 1500 m. If F_A and F_B are the forces applied by the brakes on cars A and B , respectively, then the ratio F_A/F_B is
 - $\frac{1}{3}$
 - $\frac{1}{2}$
 - $\frac{3}{2}$
 - $\frac{2}{3}$

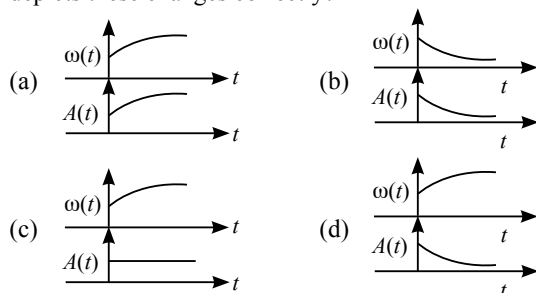
- The current passing through the battery in the given circuit, is



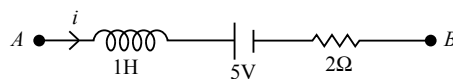
- 2.5 A
- 1.5 A
- 2.0 A
- 0.5 A

- Three identical heat conducting rods are connected in series as shown in the figure. The rods on the sides have thermal conductivity $2K$ while that in the middle has thermal conductivity K . The left end of the combination is maintained at temperature $3T$ and the right end at T . The rods are thermally insulated from outside. In steady state, temperature at the left junction is T_1 and that at the right junction is T_2 . The ratio T_1/T_2 is
 - $\frac{5}{3}$
 - $\frac{5}{4}$
 - $\frac{3}{2}$
 - $\frac{4}{3}$

- In an oscillating spring mass system, a spring is connected to a box filled with sand. As the box oscillates, sand leaks slowly out of the box vertically so that the average frequency $\omega(t)$ and average amplitude $A(t)$ of the system change with time t . Which one of the following options schematically depicts these changes correctly?



- AB is a part of an electrical circuit (see figure). The potential difference " $V_A - V_B$ ", at the instant when current $i = 2 \text{ A}$ and is increasing at a rate of 1 amp/second is



- 9 volt
- 10 volt
- 5 volt
- 6 volt

10. A particle of mass m is moving around the origin with a constant force F pulling it towards the origin. If Bohr model is used to describe its motion, the radius of the n^{th} orbit and the particle's speed v in the orbit depend on n as

(a) $r \propto n^{2/3}; v \propto n^{1/3}$ (b) $r \propto n^{4/3}; v \propto n^{-1/3}$
 (c) $r \propto n^{1/3}; v \propto n^{1/3}$ (d) $r \propto n^{1/3}; v \propto n^{2/3}$

11. In some appropriate units, time (t) and position (x) relation of a moving particle is given by $t = x^2 + x$. The acceleration of the particle is

(a) $+\frac{2}{(x+1)^3}$ (b) $+\frac{2}{2x+1}$
 (c) $-\frac{2}{(x+2)^3}$ (d) $-\frac{2}{(2x+1)^3}$

12. A model for quantized motion of an electron in a uniform magnetic field B states that the flux passing through the orbit of the electron is $n(h/e)$ where n is an integer, h is Planck's constant and e is the magnitude of electron's charge. According to the model, the magnetic moment of an electron in its lowest energy state will be (m is the mass of the electron)

(a) $\frac{heB}{\pi m}$ (b) $\frac{heB}{2\pi m}$ (c) $\frac{he}{\pi m}$ (d) $\frac{he}{2\pi m}$

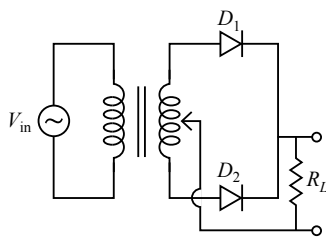
13. A microscope has an objective of focal length 2 cm, eyepiece of focal length 4 cm and the tube length of 40 cm. If the distance of distinct vision of eye is 25 cm, the magnification in the microscope is

(a) 150 (b) 250 (c) 100 (d) 125

14. There are two inclined surfaces of equal length (L) and same angle of inclination 45° with the horizontal. One of them is rough and the other is perfectly smooth. A given body takes 2 times as much time to slide down on rough surface than on the smooth surface. The coefficient of kinetic friction (μ_k) between the object and the rough surface is close to

(a) 0.5 (b) 0.75 (c) 0.25 (d) 0.40

15. A full wave rectifier circuit with diodes (D_1) and (D_2) is shown in the figure. If input supply voltage $V_{\text{in}} = 220\sin(100\pi t)$ volt, then at $t = 15$ msec



- (a) D_1 and D_2 both are forward biased
 (b) D_1 and D_2 both are reverse biased
 (c) D_1 is forward biased, D_2 is reverse biased
 (d) D_1 is reverse biased, D_2 is forward biased

16. A uniform rod of mass 20 kg and length 5 m leans against a smooth vertical wall making an angle of 60° with it. The other end rests on a rough horizontal floor. The friction force that the floor exerts on the rod is (take $g = 10 \text{ m/s}^2$)

(a) 200 N (b) $200\sqrt{3}$ N (c) 100 N (d) $100\sqrt{3}$ N

17. Two identical charged conducting spheres A and B have their centres separated by a certain distance. Charge on each sphere is q and the force of repulsion between them is F . A third identical uncharged conducting sphere is brought in contact with sphere A first and then with B and finally removed from both. New force of repulsion between spheres A and B (Radii of A and B are negligible compared to the distance of separation so that for calculating force between them they can be considered as point charges) is best given as:

(a) $\frac{F}{2}$ (b) $\frac{3F}{8}$ (c) $\frac{3F}{5}$ (d) $\frac{2F}{3}$

18. Two cities X and Y are connected by a regular bus service with a bus leaving in either direction every T min. A girl is driving scooter with a speed of 60 km/h in the direction X to Y notices that a bus goes past her every 30 minutes in the direction of her motion, and every 10 minutes in the opposite direction. Choose the correct option for the period T of the bus service and the speed (assumed constant) of the buses.

(a) 10 min, 90 km/h (b) 15 min, 120 km/h
 (c) 9 min, 40 km/h (d) 25 min, 100 km/h

19. A container has two chambers of volumes $V_1 = 2$ litres and $V_2 = 3$ litres separated by a partition made of a thermal insulator. The chambers contains $n_1 = 5$ and $n_2 = 4$ moles of ideal gas at pressures $p_1 = 1$ atm and $p_2 = 2$ atm, respectively. When the partition is removed, the mixture attains an equilibrium pressure of

(a) 1.4 atm (b) 1.8 atm (c) 1.3 atm (d) 1.6 atm

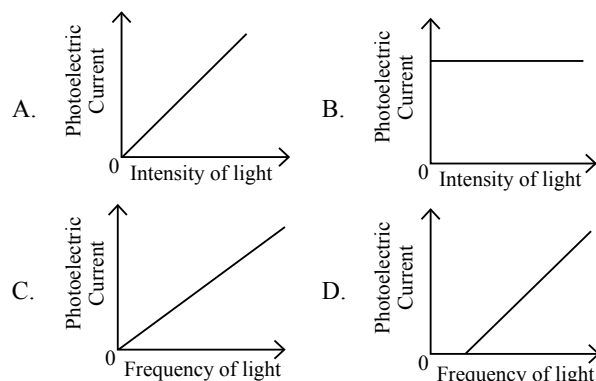
20. de-Broglie wavelength of an electron orbiting in the $n = 2$ state of hydrogen atom is close to
 (Given Bohr radius = 0.052 nm)

(a) 1.67 nm (b) 2.67 nm (c) 0.067 nm (d) 0.67 nm

21. To an ac power supply of 220 V at 50 Hz, a resistor of 20Ω , a capacitor of reactance 25Ω and an inductor of reactance 45Ω are connected in series. The corresponding current in the circuit and the phase angle between the current and the voltage is, respectively

(a) 15.6 A and 30° (b) 15.6 A and 45°
 (c) 7.8 A and 30° (d) 7.8 A and 45°

22. Which of the following options represent the variation of photoelectric current with property of light shown on the x-axis?



(a) A and D (b) B and D (c) A only (d) A and C

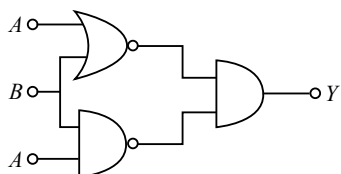
23. A pipe open at both ends has a fundamental frequency f in air. The pipe is now dipped vertically in a water drum to half of its length. The fundamental frequency of the air column is now equal to

(a) $\frac{3f}{2}$ (b) $2f$ (c) $\frac{f}{2}$ (d) f

24. Two identical point masses P and Q , suspended from two separate massless springs of spring constants k_1 and k_2 , respectively, oscillate vertically. If their maximum speeds are the same, the ratio (A_Q/A_P) of the amplitude A_Q of mass Q to the amplitude A_P of mass P is

(a) $\sqrt{\frac{k_2}{k_1}}$ (b) $\sqrt{\frac{k_1}{k_2}}$ (c) $\frac{k_2}{k_1}$ (d) $\frac{k_1}{k_2}$

25. The output (Y) of the given logic implementation is similar to the output of an/a _____ gate.



- (a) OR (b) NOR (c) AND (d) NAND

26. An oxygen cylinder of volume 30 litre has 18.20 moles of oxygen. After some oxygen is withdrawn from the cylinder, its gauge pressure drops to 11 atmospheric pressure at temperature 27°C . The mass of the oxygen withdrawn from the cylinder is nearly equal to

[Given, $R = \frac{100}{12} \text{ J mol}^{-1}\text{K}^{-1}$, and molecular mass of $\text{O}_2 = 32$, 1 atm pressure = $1.01 \times 10^5 \text{ N/m}^2$]

- (a) 0.116 kg (b) 0.156 kg (c) 0.125 kg (d) 0.144 kg

27. In a certain camera, a combination of four similar thin convex lenses are arranged axially in contact. Then the power of the combination and the total magnification in comparison to the power (p) and magnification (m) for each lens will be, respectively

- (a) $4p$ and m^4 (b) p^4 and m^4
(c) $4p$ and $4m$ (d) p^4 and $4m$

28. Two gases A and B are filled at the same pressure in separate cylinders with movable pistons of radius r_A and r_B , respectively. On supplying an equal amount of heat to both the systems reversibly under constant pressure, the pistons of gas A and B are displaced by 16 cm and 9 cm, respectively. If the change in their internal energy is the same, then the ratio $\frac{r_A}{r_B}$ is equal to

(a) $\frac{2}{\sqrt{3}}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{4}{3}$ (d) $\frac{3}{4}$

29. A balloon is made of a material of surface tension S and its inflation outlet (from where gas is filled in it) has small area A . It is filled with a gas of density ρ and takes a spherical shape of radius R . When the gas is allowed to flow freely out of it, its radius r changes from R to 0 (zero) in time T . If the speed

$v(r)$ of gas coming out of the balloon depends on r as r^a and $T \propto S^\alpha A^\beta \rho^\gamma R^\delta$ then

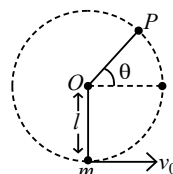
(a) $a = -\frac{1}{2}, \alpha = -\frac{1}{2}, \beta = -1, \gamma = \frac{1}{2}, \delta = \frac{7}{2}$

(b) $a = \frac{1}{2}, \alpha = \frac{1}{2}, \beta = -\frac{1}{2}, \gamma = \frac{1}{2}, \delta = \frac{7}{2}$

(c) $a = \frac{1}{2}, \alpha = \frac{1}{2}, \beta = -1, \gamma = +1, \delta = \frac{3}{2}$

(d) $a = -\frac{1}{2}, \alpha = -\frac{1}{2}, \beta = -1, \gamma = -\frac{1}{2}, \delta = \frac{5}{2}$

30. A bob of heavy mass m is suspended by a light string of length l . The bob is given a horizontal velocity v_0 as shown in figure. If the string gets slack at some point P making an angle θ from the horizontal, the ratio of the speed v of the bob at point P to its initial speed v_0 is



(a) $\left(\frac{\cos \theta}{2 + 3 \sin \theta}\right)^{1/2}$ (b) $\left(\frac{\sin \theta}{2 + 3 \sin \theta}\right)^{1/2}$

(c) $(\sin \theta)^{1/2}$ (d) $\left(\frac{1}{2 + 3 \sin \theta}\right)^{1/2}$

31. A physical quantity P is related to four observations a, b, c and d as follows:

$$P = a^3 b^2 / c \sqrt{d}$$

The percentage errors of measurement in a, b, c and d are 1%, 3%, 2% and 4% respectively. The percentage error in the quantity P is

- (a) 13% (b) 15% (c) 10% (d) 2%

32. The Sun rotates around its centre once in 27 days. What will be the period of revolution if the Sun were to expand to twice its present radius without any external influence? Assume the Sun to be a sphere of uniform density.

- (a) 115 days (b) 108 days (c) 100 days (d) 105 days

33. The radius of Martian orbit around the Sun is about 4 times the radius of the orbit of Mercury. The Martian year is 687 Earth days. Then which of the following is the length of 1 year on Mercury?

- (a) 172 earth days (b) 124 earth days
(c) 88 earth days (d) 225 earth days

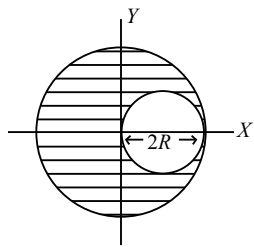
34. A wire of resistance R is cut into 8 equal pieces. From these pieces two equivalent resistances are made by adding four of these together in parallel. Then these two sets are added in series. The net effective resistance of the combination is

(a) $\frac{R}{16}$ (b) $\frac{R}{8}$ (c) $\frac{R}{64}$ (d) $\frac{R}{32}$

35. A photon and an electron (mass m) have the same energy E . The ratio $(\lambda_{\text{photon}}/\lambda_{\text{electron}})$ of their de-Broglie wavelengths is (c is the speed of light)

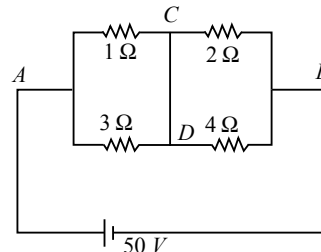
(a) $c\sqrt{\frac{2m}{E}}$ (b) $\frac{1}{c}\sqrt{\frac{E}{2m}}$ (c) $\sqrt{\frac{E}{2m}}$ (d) $c\sqrt{2mE}$

36. A sphere of radius R is cut from a larger solid sphere of radius $2R$ as shown in the figure. The ratio of the moment of inertia of the smaller sphere to that of the rest part of the sphere about the Y -axis is

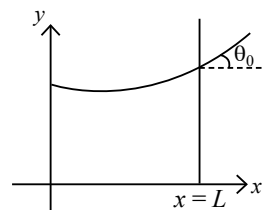


- (a) $\frac{7}{57}$ (b) $\frac{7}{64}$ (c) $\frac{7}{8}$ (d) $\frac{7}{40}$
37. An electron (mass 9×10^{-31} kg and charge 1.6×10^{-19} C) moving with speed $c/100$ (c = speed of light) is injected into a magnetic field $\vec{B}$ of magnitude 9×10^{-4} T perpendicular to its direction of motion. We wish to apply an uniform electric field $\vec{E}$ together with the magnetic field so that the electron does not deflect from its path. Then
(speed of light $c = 3 \times 10^8$ m s $^{-1}$)
- (a) $\vec{E}$ is parallel to $\vec{B}$ and its magnitude is 27×10^2 V m $^{-1}$
 (b) $\vec{E}$ is parallel to $\vec{B}$ and its magnitude is 27×10^4 V m $^{-1}$
 (c) $\vec{E}$ is perpendicular to $\vec{B}$ and its magnitude is 27×10^4 V m $^{-1}$
 (d) $\vec{E}$ is perpendicular to $\vec{B}$ and its magnitude is 27×10^2 V m $^{-1}$
38. The electric field in a plane electromagnetic wave is given by $E_z = 60 \cos(5x + 1.5 \times 10^9 t)$ V/m. Then expression for the corresponding magnetic field is (here subscripts denote the direction of the field)
- (a) $B_z = 60 \cos(5x + 1.5 \times 10^9 t)$ T
 (b) $B_y = 60 \sin(5x + 1.5 \times 10^9 t)$ T
 (c) $B_y = 2 \times 10^{-7} \cos(5x + 1.5 \times 10^9 t)$ T
 (d) $B_x = 2 \times 10^{-7} \cos(5x + 1.5 \times 10^9 t)$ T
39. A body weighs 48 N on the surface of the earth. The gravitational force experienced by the body due to the earth at a height equal to one-third the radius of the earth from its surface is
- (a) 32 N (b) 36 N (c) 16 N (d) 27 N
40. An unpolarized light beam travelling in air is incident on a medium of refractive index 1.73 at Brewster's angle. Then
- (a) both reflected and transmitted light are perfectly polarized with angles of reflection and refraction close to 60° and 30° , respectively.
 (b) transmitted light is completely polarized with angle of refraction close to 30°
 (c) reflected light is completely polarized and the angle of reflection is close to 60°
 (d) reflected light is partially polarized and the angle of reflection is close to 30°

41. A constant voltage of 50 V is maintained between the points A and B of the circuit shown in the figure. The current through the branch CD of the circuit is



- (a) 2.5 A (b) 3.0 A (c) 1.5 A (d) 2.0 A
42. The plates of a parallel plate capacitor are separated by d . Two slabs of different dielectric constant K_1 and K_2 with thickness $\frac{3}{8}d$ and $\frac{d}{2}$, respectively are inserted in the capacitor. Due to this, the capacitance becomes two times larger than when there is nothing between the plates. If $K_1 = 1.25 K_2$, the value of K_1 is
- (a) 1.60 (b) 1.33 (c) 2.66 (d) 2.33
43. Consider the diameter of a spherical object being measured with the help of a Vernier callipers. Suppose its 10 Vernier Scale Divisions (V.S.D.) are equal to its 9 Main Scale Divisions (M.S.D.). The least division in the M.S. is 0.1 cm and the zero of V.S. is at $x = 0.1$ cm when the jaws of Vernier callipers are closed.
- If the main scale reading for the diameter is $M = 5$ cm and the number of coinciding vernier division is 8, the measured diameter after zero error correction, is
- (a) 4.98 cm (b) 5.00 cm (c) 5.18 cm (d) 5.08 cm
44. A 2 amp current is flowing through two different small circular copper coils having radii ratio 1:2. The ratio of their respective magnetic moments will be
- (a) 2:1 (b) 4:1 (c) 1:4 (d) 1:2
45. Consider a water tank shown in the figure. It has one wall at $x = L$ and can be taken to be very wide in the z direction. When filled with a liquid of surface tension S and density ρ , the liquid surface makes angle θ_0 ($\theta_0 \ll 1$) with the x -axis at $x = L$. If $y(x)$ is the height of the surface then the equation for $y(x)$ is



(Take $\theta(x) = \sin \theta(x) = \tan \theta(x) = \frac{dy}{dx}$, g is the acceleration due to gravity)

- (a) $\frac{d^2 y}{dx^2} = \sqrt{\frac{\rho g}{S}}$ (b) $\frac{dy}{dx} = \sqrt{\frac{\rho g}{S}} x$
 (c) $\frac{d^2 y}{dx^2} = \frac{\rho g}{S} x$ (d) $\frac{d^2 y}{dx^2} = \frac{\rho g}{S} y$

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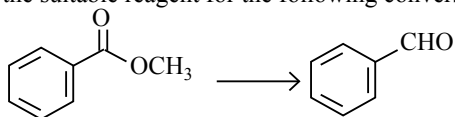
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CHEMISTRY

46. Identify the suitable reagent for the following conversion.



- (a) (i) NaBH_4 , (ii) $\text{H}^+/\text{H}_2\text{O}$ (b) $\text{H}_2/\text{Pd}-\text{BaSO}_4$
 (c) (i) LiAlH_4 , (ii) $\text{H}^+/\text{H}_2\text{O}$ (d) (i) $\text{AlH}(\text{iBu})_2$, (ii) H_2O
47. The correct order of decreasing acidity of the following aliphatic acids is
 (a) $\text{HCOOH} > \text{CH}_3\text{COOH} > (\text{CH}_3)_2\text{CHCOOH} > (\text{CH}_3)_3\text{CCOOH}$
 (b) $\text{HCOOH} > (\text{CH}_3)_3\text{CCOOH} > (\text{CH}_3)_2\text{CHCOOH} > \text{CH}_3\text{COOH}$
 (c) $(\text{CH}_3)_3\text{CCOOH} > (\text{CH}_3)_2\text{CHCOOH} > \text{CH}_3\text{COOH} > \text{HCOOH}$
 (d) $\text{CH}_3\text{COOH} > (\text{CH}_3)_2\text{CHCOOH} > (\text{CH}_3)_3\text{CCOOH} > \text{HCOOH}$
48. Which one of the following reactions does not belong to "Lassaigne's test"?
 (a) $\text{Na} + \text{X} \xrightarrow{\Delta} \text{NaX}$
 (b) $2\text{CuO} + \text{C} \xrightarrow{\Delta} 2\text{Cu} + \text{CO}_2$
 (c) $\text{Na} + \text{C} + \text{N} \xrightarrow{\Delta} \text{NaCN}$
 (d) $2\text{Na} + \text{S} \xrightarrow{\Delta} \text{Na}_2\text{S}$
49. If the rate constant of a reaction is 0.03 s^{-1} , how much time does it take for 7.2 mol L^{-1} concentration of the reactant to get reduced to 0.9 mol L^{-1} ?
 (Given : $\log 2 = 0.301$)
 (a) 210 s (b) 21.0 s (c) 69.3 s (d) 23.1 s
50. Given below are two statements :
Statement I : A hypothetical diatomic molecule with bond order zero is quite stable.
Statement II : As bond order increases, the bond length increases.
 In the light of the above statements, choose the most appropriate answer from the options given below:
 (a) Statement I is true but Statement II is false.
 (b) Statement I is false but Statement II is true.
 (c) Both Statement I and Statement II are true.
 (d) Both Statement I and Statement II are false.
51. Out of the following complex compounds, which of the compound will be having the minimum conductance in solution?
 (a) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (b) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}$
 (c) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ (d) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$
52. Which of the following aqueous solution will exhibit highest boiling point?
 (a) 0.01 M Na_2SO_4 (b) 0.015 M $\text{C}_6\text{H}_{12}\text{O}_6$
 (c) 0.01 M urea (d) 0.01 M KNO_3
53. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : I undergoes $\text{S}_{\text{N}}2$ reaction faster than Cl.

Reason (R) : Iodine is a better leaving group because of its large size.

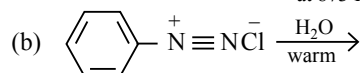
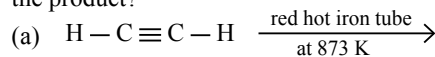
In the light of the above statements, choose the correct answer from the options given below:

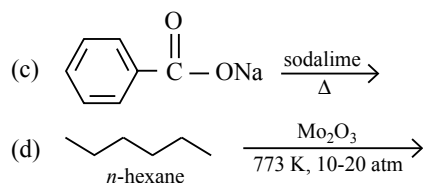
- (a) A is true but R is false.
 (b) A is false but R is true.
 (c) Both A and R are true and R is the correct explanation of A.
 (d) Both A and R are true but R is not the correct explanation of A.
54. Consider the following compounds:
 $\text{K}\underline{\text{O}}_2$, $\text{H}_2\underline{\text{O}}_2$ and $\text{H}_2\underline{\text{S}}\text{O}_4$
 The oxidation states of the underlined elements in them are, respectively,
 (a) +1, -2 and +4 (b) +4, -4 and +6
 (c) +1, -1 and +6 (d) +2, -2 and +6
55. Match List-I with List-II.

List-I		List-II	
A.	Haber process	I.	Fe catalyst
B.	Wacker oxidation	II.	PdCl_2
C.	Wilkinson catalyst	III.	$[(\text{PPh}_3)_3\text{RhCl}]$
D.	Ziegler catalyst	IV.	TiCl_4 with $\text{Al}(\text{CH}_3)_3$

Choose the correct answer from the options given below:

- (a) A-I, B-II, C-III, D-IV
 (b) A-I, B-IV, C-III, D-II
 (c) A-I, B-II, C-IV, D-III
 (d) A-II, B-III, C-I, D-IV
56. Given below are two statements :
Statement I : Like nitrogen that can form ammonia, arsenic can form arsine.
Statement II : Antimony cannot form antimony pentoxide.
 In the light of the above statements, choose the most appropriate answer from the options given below:
 (a) Statement I is correct but Statement II is incorrect.
 (b) Statement I is incorrect but Statement II is correct.
 (c) Both Statement I and Statement II are correct.
 (d) Both Statement I and Statement II are incorrect.
57. Given below are two statements :
Statement I : Ferromagnetism is considered as an extreme form of paramagnetism.
Statement II : The number of unpaired electrons in a Cr^{2+} ion ($Z = 24$) is the same as that of a Nd^{3+} ion ($Z = 60$).
 In the light of the above statements, choose the correct answer from the options given below:
 (a) Statement I is true but Statement II is false.
 (b) Statement I is false but Statement II is true.
 (c) Both Statement I and Statement II are true.
 (d) Both Statement I and Statement II are false.
58. Which one of the following reactions does not give benzene as the product?





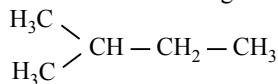
59. Match List-I with List-II.

List-I		List-II	
A.	XeO ₃	I.	sp^3d ; linear
B.	XeF ₂	II.	sp^3 ; pyramidal
C.	XeOF ₄	III.	sp^3d^3 , distorted octahedral
D.	XeF ₆	IV.	sp^3d^2 , square pyramidal

Choose the correct answer from the options given below:

- (a) A-IV, B-II, C-III, D-I
 (b) A-IV, B-II, C-I, D-III
 (c) A-II, B-I, C-IV, D-III
 (d) A-II, B-I, C-III, D-IV

60. How many products (including stereoisomers) are expected from monochlorination of the following compound?



- (a) 5 (b) 6 (c) 2 (d) 3

61. Which of the following statements are true?

- A. Unlike Ga that has a very high melting point, Cs has a very low melting point.
 B. On Pauling scale, the electronegativity values of N and Cl are not the same.
 C. Ar, K⁺, Cl⁻, Ca²⁺ and S²⁻ are all isoelectronic species.
 D. The correct order of the first ionization enthalpies of Na, Mg, Al and Si is Si > Al > Mg > Na.
 E. The atomic radius of Cs is greater than that of Li and Rb.

Choose the correct answer from the options given below:

- (a) C and D only (b) A, C and E only
 (c) A, B and E only (d) C and E only

62. The standard heat of formation in kcal/mol of Ba²⁺ is [Given: standard heat of formation of SO₄²⁻ ion_(aq) = -216 kcal/mol, standard heat of crystallisation of BaSO_{4(s)} = -4.5 kcal/mol, standard heat of formation of BaSO_{4(s)} = -349 kcal/mol]
 (a) +133.0 (b) +220.5 (c) -128.5 (d) -133.0

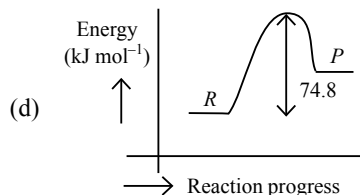
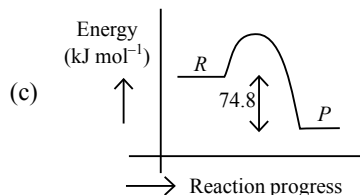
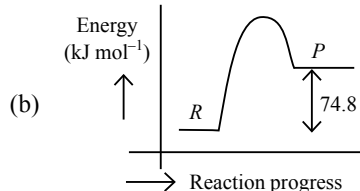
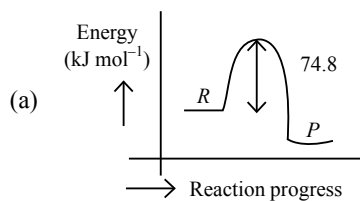
63. Match List-I with List-II.

List-I (Example)		List-II (Type of Solution)	
A.	Humidity	I.	Solid in solid
B.	Alloys	II.	Liquid in gas
C.	Amalgams	III.	Solid in gas
D.	Smoke	IV.	Liquid in solid

Choose the correct answer from the options given below:

- (a) A-III, B-I, C-IV, D-II (b) A-III, B-II, C-I, D-IV
 (c) A-II, B-IV, C-I, D-III (d) A-II, B-I, C-IV, D-III

64. $\text{C}_{(s)} + 2\text{H}_{2(g)} \rightarrow \text{CH}_{4(g)}$; $\Delta H = -74.8 \text{ kJ mol}^{-1}$
 Which of the following diagrams gives an accurate representation of the above reaction?
 [R → reactants, P → products]



65. Sugar 'X'

- A. is found in honey.
 B. is a keto sugar.
 C. exists in α and β -anomeric forms.
 D. is laevorotatory.

'X' is

- (a) maltose (b) sucrose
 (c) D-glucose (d) D-fructose

66. Total number of possible isomers (both structural as well as stereoisomers) of cyclic ethers of molecular formula C₄H₈O is
 (a) 10 (b) 11 (c) 6 (d) 8

67. For the reaction $A_{(g)} \rightleftharpoons 2B_{(g)}$, the backward reaction rate constant is higher than the forward reaction rate constant by a factor of 2500, at 1000 K.

[Given: $R = 0.0831 \text{ L atm mol}^{-1} \text{ K}^{-1}$]

K_p for the reaction at 1000 K is

- (a) 0.033 (b) 0.021
 (c) 83.1 (d) 2.077×10^5

68. The ratio of the wavelengths of the light absorbed by a hydrogen atom when it undergoes $n = 2 \rightarrow n = 3$ and $n = 4 \rightarrow n = 6$ transitions, respectively, is

- (a) $\frac{1}{9}$ (b) $\frac{1}{4}$ (c) $\frac{1}{36}$ (d) $\frac{1}{16}$

69. If the molar conductivity (Λ_m) of a 0.050 mol L⁻¹ solution of a monobasic weak acid is 90 S cm² mol⁻¹, its extent (degree) of dissociation will be

[Assume $\Lambda_+^\circ = 349.6 \text{ S cm}^2 \text{ mol}^{-1}$ and $\Lambda_-^\circ = 50.4 \text{ S cm}^2 \text{ mol}^{-1}$]

- (a) 0.225 (b) 0.215 (c) 0.115 (d) 0.125

70. 5 moles of liquid X and 10 moles of liquid Y make a solution having a vapour pressure of 70 torr. The vapour pressures of pure X and Y are 63 torr and 78 torr respectively. Which of the following is true regarding the described solution?

(a) The solution is ideal.
 (b) The solution has volume greater than the sum of individual volumes.
 (c) The solution shows positive deviation.
 (d) The solution shows negative deviation.

71. Among the following, choose the ones with equal number of atoms.

A. 212 g of $\text{Na}_2\text{CO}_{3(s)}$ [molar mass = 106 g]
 B. 248 g of $\text{Na}_2\text{O}_{(s)}$ [molar mass = 62 g]
 C. 240 g of $\text{NaOH}_{(s)}$ [molar mass = 40 g]
 D. 12 g of $\text{H}_{2(g)}$ [molar mass = 2 g]
 E. 220 g of $\text{CO}_{2(g)}$ [molar mass = 44 g]

Choose the correct answer from the options given below :

(a) B, C, and D only (b) B, D, and E only
 (c) A, B, and C only (d) A, B, and D only

72. Which of the following are paramagnetic?

A. $[\text{NiCl}_4]^{2-}$ B. $\text{Ni}(\text{CO})_4$
 C. $[\text{Ni}(\text{CN})_4]^{2-}$ D. $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
 E. $\text{Ni}(\text{PPh}_3)_4$

Choose the correct answer from the options given below :

(a) A and D only (b) A, D and E only
 (c) A and C only (d) B and E only.

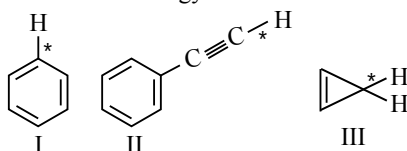
73. If the half-life ($t_{1/2}$) for a first order reaction is 1 minute, then the time required for 99.9% completion of the reaction is closest to

(a) 5 minutes (b) 10 minutes
 (c) 2 minutes (d) 4 minutes.

74. Energy and radius of first Bohr orbit of He^+ and Li^{2+} are [Given $R_H = 2.18 \times 10^{-18} \text{ J}$, $a_0 = 52.9 \text{ pm}$]

(a) $E_n(\text{Li}^{2+}) = -19.62 \times 10^{-16} \text{ J}$; $r_n(\text{Li}^{2+}) = 17.6 \text{ pm}$
 $E_n(\text{He}^+) = -8.72 \times 10^{-16} \text{ J}$; $r_n(\text{He}^+) = 26.4 \text{ pm}$
 (b) $E_n(\text{Li}^{2+}) = -8.72 \times 10^{-16} \text{ J}$; $r_n(\text{Li}^{2+}) = 17.6 \text{ pm}$
 $E_n(\text{He}^+) = -19.62 \times 10^{-16} \text{ J}$; $r_n(\text{He}^+) = 17.6 \text{ pm}$
 (c) $E_n(\text{Li}^{2+}) = -19.62 \times 10^{-18} \text{ J}$; $r_n(\text{Li}^{2+}) = 17.6 \text{ pm}$
 $E_n(\text{He}^+) = -8.72 \times 10^{-18} \text{ J}$; $r_n(\text{He}^+) = 26.4 \text{ pm}$
 (d) $E_n(\text{Li}^{2+}) = -8.72 \times 10^{-18} \text{ J}$; $r_n(\text{Li}^{2+}) = 26.4 \text{ pm}$
 $E_n(\text{He}^+) = -19.62 \times 10^{-18} \text{ J}$; $r_n(\text{He}^+) = 17.6 \text{ pm}$

75. Among the given compounds I-III, the correct order of bond dissociation energy of C—H bond marked with * is



(a) III > II > I (b) II > III > I
 (c) II > I > III (d) I > II > III

76. Dalton's atomic theory could not explain which of the following?

(a) Law of multiple proportion
 (b) Law of gaseous volume
 (c) Law of conservation of mass
 (d) Law of constant proportion

77. Identify the correct orders against the property mentioned.

A. $\text{H}_2\text{O} > \text{NH}_3 > \text{CHCl}_3$ - Dipole moment
 B. $\text{XeF}_4 > \text{XeO}_3 > \text{XeF}_2$ - Number of lone pairs on central atom
 C. $\text{O} - \text{H} > \text{C} - \text{H} > \text{N} - \text{O}$ - Bond length
 D. $\text{N}_2 > \text{O}_2 > \text{H}_2$ - Bond enthalpy

Choose the correct answer from the options given below :

(a) A, C only (b) B, C only
 (c) A, D only (d) B, D only

78. Match List I with List II.

List I (Name of Vitamin)		List II (Deficiency disease)	
A.	Vitamin B ₁₂	I.	Cheilosis
B.	Vitamin D	II.	Convulsions
C.	Vitamin B ₂	III.	Rickets
D.	Vitamin B ₆	IV.	Pernicious anaemia

Choose the correct answer from the options given below :

(a) A-II, B-III, C-I, D-IV (b) A-IV, B-III, C-II, D-I
 (c) A-I, B-III, C-II, D-IV (d) A-IV, B-III, C-I, D-II

79. The correct order of decreasing basic strength of the given amines is

(a) N -ethylethanamine > ethanamine > N -methylaniline > benzenamine
 (b) benzenamine > ethanamine > N -methylaniline > N -ethylethanamine
 (c) N -methylaniline > benzenamine > ethanamine > N -ethylethanamine
 (d) N -ethylethanamine > ethanamine > p benzenamine > N -methylaniline.

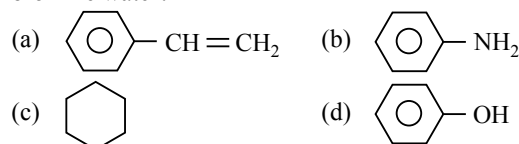
80. The correct order of the wavelength of light absorbed by the following complexes is,

A. $[\text{Co}(\text{NH}_3)_6]^{3+}$ B. $[\text{Co}(\text{CN})_6]^{3-}$
 C. $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ D. $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$

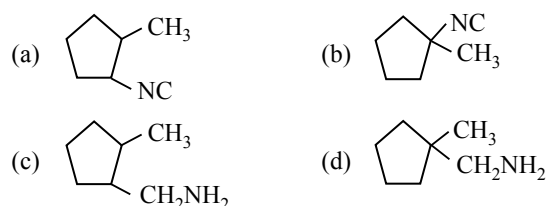
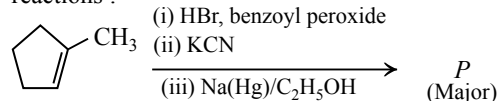
Choose the correct answer from the options given below :

(a) $\text{C} < \text{D} < \text{A} < \text{B}$ (b) $\text{C} < \text{A} < \text{D} < \text{B}$
 (c) $\text{B} < \text{D} < \text{A} < \text{C}$ (d) $\text{B} < \text{A} < \text{D} < \text{C}$

81. Which one of the following compounds does not decolourize bromine water?



82. Predict the major product 'P' in the following sequence of reactions :



83. Match List I with List II.

List I (Mixture)		List II (Method of separation)	
A.	$\text{CHCl}_3 + \text{C}_6\text{H}_5\text{NH}_2$	I.	Distillation under reduced pressure
B.	Crude oil in petroleum industry	II.	Steam distillation
C.	Glycerol from spent-lye	III.	Fractional distillation
D.	Aniline-water	IV.	Simple distillation

Choose the correct answer from the options given below :

- (a) A-III, B-IV, C-I, D-II (b) A-III, B-IV, C-II, D-I
(c) A-IV, B-III, C-I, D-II (d) A-IV, B-III, C-II, D-I

84. Which among the following electronic configurations belong to main group elements?

- A. $[\text{Ne}]3s^1$ B. $[\text{Ar}]3d^34s^2$
C. $[\text{Kr}]4d^{10}5s^25p^5$ D. $[\text{Ar}]3d^{10}4s^1$
E. $[\text{Rn}]5f^06d^27s^2$

Choose the correct answer from the option given below :

- (a) D and E only (b) A, C and D only
(c) B and E only (d) A and C only

85. Which one of the following compounds can exist as *cis-trans* isomers?

- (a) 1,1-Dimethylcyclopropane
(b) 1,2-Dimethylcyclohexane
(c) Pent-1-ene
(d) 2-Methylhex-2-ene

86. Phosphoric acid ionizes in three steps with their ionization constant values K_{a_1} , K_{a_2} , and K_{a_3} , respectively, while K is the overall ionization constant.

Which of the following statements are true?

- A. $\log K = \log K_{a_1} + \log K_{a_2} + \log K_{a_3}$
B. H_3PO_4 is a stronger acid than H_2PO_4^- and HPO_4^{2-} .
C. $K_{a_1} > K_{a_2} > K_{a_3}$
D. $K_{a_1} = \frac{K_{a_3} + K_{a_2}}{2}$

Choose the correct answer from the options given below :

- (a) B, C and D only (b) A, B and C only
(c) A and B only (d) A and C only

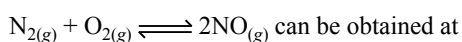
87. Match List I with List II.

List I (Ion)		List II (Group number in cation analysis)	
A.	Co^{2+}	I.	Group-I
B.	Mg^{2+}	II.	Group-III
C.	Pb^{2+}	III.	Group-IV
D.	Al^{3+}	IV.	Group-VI

Choose the correct answer from the options given below :

- (a) A-III, B-II, C-IV, D-I (b) A-III, B-II, C-I, D-IV
(c) A-III, B-IV, C-II, D-I (d) A-III, B-IV, C-I, D-II

88. Higher yield of NO in

[ΔH of the reaction = $+180.7 \text{ kJ mol}^{-1}$]

- A. higher temperature
B. lower temperature
C. higher concentration of N_2
D. higher concentration of O_2 .

Choose the correct answer from the options given below :

- (a) B, C, D only (b) A, C, D only
(c) A, D only (d) B, C only

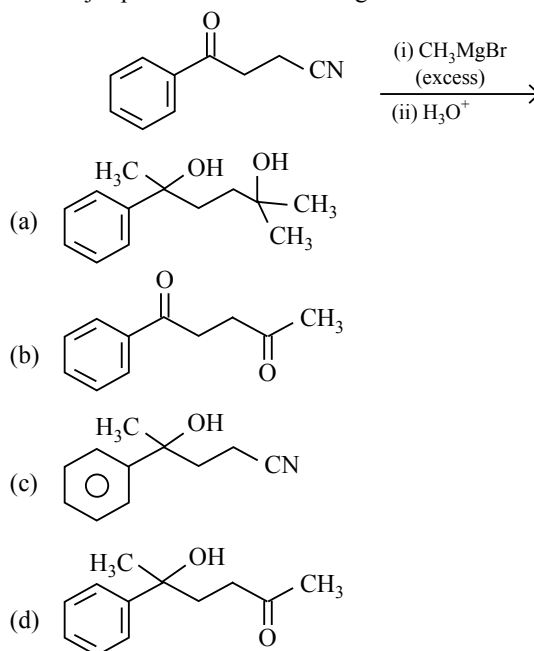
89. Given below are two statements :

Statement I : Benzenediazonium salt is prepared by the reaction of aniline with nitrous acid at 273-278 K. It decomposes easily in the dry state.**Statement II :** Insertion of iodine into the benzene ring is difficult and hence iodobenzene is prepared through the reaction of benzenediazonium salt with KI.

In the light of the above statements, choose the most appropriate answer from the options given below:

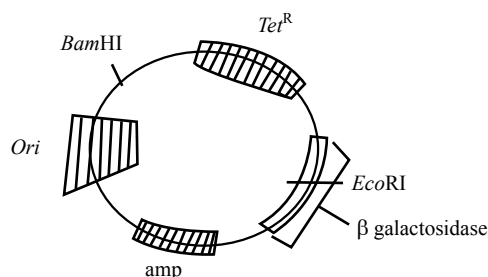
- (a) Statement I is correct but Statement II is incorrect.
(b) Statement I is incorrect but Statement II is correct.
(c) Both Statement I and Statement II are correct.
(d) Both Statement I and Statement II are incorrect.

90. The major product of the following reaction is



BIOLOGY

91.



In the above represented plasmid an alien piece of DNA is inserted at *EcoRI* site. Which of the following strategies will be chosen to select the recombinant colonies?

- White color colonies will be selected.
- Blue color colonies grown on ampicillin plates can be selected.
- Using ampicillin & tetracyclin containing medium plate.
- Blue color colonies will be selected.

92. The protein portion of an enzyme is called

- apoenzyme
- prosthetic group
- cofactor
- coenzyme.

93. Given below are two statements:

Statement I: The primary source of energy in an ecosystem is solar energy.

Statement II: The rate of production of organic matter during photosynthesis in an ecosystem is called net primary productivity (NPP).

In the light of the above statements, choose the most appropriate answer from the options given below:

- Statement I is correct but statement II is incorrect.
- Statement I is incorrect but statement II is correct.
- Both statement I and statement II are correct.
- Both statement I and statement II are incorrect.

94. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): A typical unfertilised, angiosperm embryo sac at maturity is 8 nucleate and 7-celled.

Reason (R): The egg apparatus has 2 polar nuclei.

In the light of the above statements, choose the correct answer from the options given below:

- A is true but R is false.
- A is false but R is true.
- Both A and R are true and R is the correct explanation of A.
- Both A and R are true but R is not the correct explanation of A.

95. Neoplastic characteristics of cells refer to:

- A mass of proliferating cell
- Rapid growth of cells
- Invasion and damage to the surrounding tissue
- Those confined to original location

Choose the correct answer from the options given below :

- A, B, D only
- B, C, D only
- A, B only
- A, B, C only.

96. Which one of the following is the characteristic feature of gymnosperms?

- Seeds are absent.
- Gymnosperms have flowers for reproduction.
- Seeds are enclosed in fruits.
- Seeds are naked.

97. Match List-I with List-II.

List-I	List-II
A. Progesterone	I. Pars intermedia
B. Relaxin	II. Ovary
C. Melanocyte stimulating hormone	III. Adrenal medulla
D. Catecholamines	IV. Corpus luteum

Choose the correct answer from the options given below.

- A-II, B-IV, C-I, D-III
- A-III, B-II, C-IV, D-I
- A-IV, B-II, C-I, D-III
- A-IV, B-II, C-III, D-I

98. Which chromosome in the human genome has the highest number of genes?

- Chromosome 1
- Chromosome 10
- Chromosome X
- Chromosome Y

99. Which of the following statements about RuBisCO is true?

- It is an enzyme involved in the photolysis of water.
- It catalyzes the carboxylation of RuBP.
- It is active only in the dark.
- It has higher affinity for oxygen than carbon dioxide.

100. The first menstruation is called

- diapause
- ovulation
- menopause
- menarche.

101. Which of the following genetically engineered organisms was used by Eli Lilly to prepare human insulin?

- Virus
- Phage
- Bacterium
- Yeast

102. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): All vertebrates are chordates but all chordates are not vertebrate.

Reason (R): The members of subphylum vertebrata possess notochord during the embryonic period, the notochord is replaced by a cartilaginous or bony vertebral column in adults. In the light of the above statements, choose the correct answer from the options given below :

- A is true but R is false.
- A is false but R is true.
- Both A and R are true and R is the correct explanation of A.
- Both A and R are true but R is not the correct explanation of A.

103. What is the main function of the spindle fibers during mitosis?

- To repair damaged DNA
- To regulate cell growth
- To separate the chromosomes
- To synthesize new DNA

104. Match List I with List II.

List-I	List-II
A. Alfred Hershey and Martha Chase	I. <i>Streptococcus pneumoniae</i>
B. Euchromatin	II. Densely packed and dark-stained
C. Frederick Griffith	III. Loosely packed and light-stained
D. Heterochromatin	IV. DNA as genetic material confirmation

Choose the correct answer from the options given below.

- A-IV, B-III, C-I, D-II
- A-III, B-II, C-IV, D-I
- A-II, B-IV, C-I, D-III
- A-IV, B-II, C-I, D-III

105. Match List I with List II.

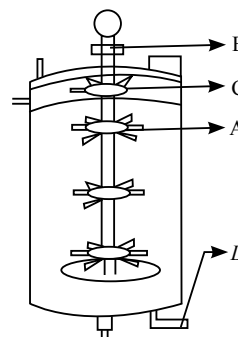
List I	List II
A. Adenosine	I. Nitrogen base
B. Adenylic acid	II. Nucleotide

- C. Adenine III. Nucleoside
D. Alanine IV. Amino acid

Choose the option with all correct matches.

- (a) A-III, B-II, C-I, D-IV (b) A-II, B-III, C-I, D-IV
(c) A-III, B-IV, C-II, D-I (d) A-III, B-II, C-IV, D-I
- 106.** In frog, the renal portal system is a special venous connection that acts to link
(a) kidney and intestine
(b) kidney and lower part of body
(c) liver and intestine
(d) liver and kidney.
- 107.** Which of the following are the post-transcriptional events in an eukaryotic cell?
A. Transport of pre-mRNA to cytoplasm prior to splicing.
B. Removal of introns and joining of exons.
C. Addition of methyl group at 5' end of hnRNA.
D. Addition of adenine residues at 3' end of hnRNA.
E. Base pairing of two complementary RNAs.
Choose the correct answer from the options given below.
(a) B, C, E only (b) C, D, E only
(c) A, B, C only (d) B, C, D only
- 108.** Polymerase chain reaction (PCR) amplifies DNA following the equation.
(a) $2n + 1$ (b) $2N^2$ (c) N^2 (d) 2^n
- 109.** Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Both wind and water pollinated flowers are not very colourful and do not produce nectar.
Reason (R): The flowers produce enormous amount of pollen grains in wind and water pollinated flowers.
In the light of the above statements, choose the correct answer from the options given below:
(a) A is true but R is false.
(b) A is false but R is true.
(c) Both A and R are true and R is the correct explanation of A.
(d) Both A and R are true but R is not the correct explanation of A.
- 110.** Epiphytes that are growing on a mango branch is an example of which of the following?
(a) Predation (b) Amensalism
(c) Commensalism (d) Mutualism
- 111.** Find the correct statements :
A. In human pregnancy, the major organ systems are formed at the end of 12 weeks.
B. In human pregnancy the major organ systems are formed at the end of 8 weeks.
C. In human pregnancy heart is formed after one month of gestation.
D. In human pregnancy, limbs and digits develop by the end of second month.
E. In human pregnancy the appearance of hair is usually observed in the fifth month.
Choose the correct answer from the options given below.
(a) B, C, D and E Only (b) A, C, D and E Only
(c) A and E Only (d) B and C Only

- 112.** Identify the part of a bio-reactor which is used as a foam braker from the given figure.



- (a) D (b) C (c) A (d) B
- 113.** Frogs respire in water by skin and buccal cavity and on land by skin, buccal cavity and lungs.
Choose the correct answer from the following:
(a) The statement is false for water but true for land.
(b) The statement is false for both the environment.
(c) The statement is true for water but false for land.
(d) The statement is true for both the environment.
- 114.** Consider the following statements regarding function of adrenaly medullary hormones:
A. It causes pupillary constriction.
B. It is a hyperglycemic hormone.
C. It causes piloerection.
D. It increases strength of heart contraction.
Choose the correct answer from the options given below:
(a) A, C and D only (b) D only
(c) C and D only (d) B, C and D only.
- 115.** Read the following statements on plant growth and development.
A. Parthenocarpy can be induced by auxins.
B. Plant growth regulators can be involved in promotion as well as inhibition of growth.
C. Dedifferentiation is a pre-requisite for re-differentiation.
D. Absciscic acid is a plant growth promoter.
E. Apical dominance promotes the growth of lateral buds.
Choose the option with all correct statements.
(a) A, D, E only (b) B, D, E only
(c) A, B, C only (d) A, C, E only
- 116.** Which of the following hormones released from the pituitary is actually synthesized in the hypothalamus?
(a) Follicle-stimulating hormone (FSH)
(b) Adenocorticotrophic hormone (ACTH)
(c) Luteinizing hormone (LH)
(d) Anti-diuretic hormone (ADH)
- 117.** Which of the following is an example of non-distilled alcoholic beverage produced by yeast?
(a) Beer (b) Rum (c) Whisky (d) Brandy
- 118.** What is the pattern of inheritance for polygenic trait?
(a) Autosomal dominant pattern
(b) X-linked recessive inheritance pattern
(c) Mendelian inheritance pattern
(d) Non-mendelian inheritance pattern

119. Match List-I with List-II.

List-I	List-II
A. Head	I. Enzymes
B. Middle piece	II. Sperm motility
C. Acrosome	III. Energy
D. Tail	IV. Genetic material

Choose the correct answer from the options given below:

- (a) A-III, B-IV, C-II, D-I (b) A-III, B-II, C-I, D-IV
(c) A-IV, B-III, C-I, D-II (d) A-IV, B-III, C-II, D-I

120. Which of the following is an example of a zygomorphic flower?

- (a) Pea (b) Chilli (c) *Petunia* (d) *Datura*

121. Which of following organisms cannot fix nitrogen?

- A. *Azotobacter* B. *Oscillatoria*
C. *Anabaena* D. *Volvox*
E. *Nostoc*

Choose the correct answer from the options given below:

- (a) B only (b) E only (c) A only (d) D only.

122. Which one of the following is an example of *ex-situ* conservation?

- (a) Zoos and botanical gardens
(b) Protected areas
(c) National Park
(d) Wildlife Sanctuary

123. Who is known as the father of Ecology in India?

- (a) Ram Udar (b) Birbal Sahni
(c) S.R. Kashyap (d) Ramdeo Misra

124. Given below are two statements:

Statement I: In the RNA world, RNA is considered the first genetic material evolved to carry out essential life processes. RNA acts as a genetic material and also as a catalyst for some important biochemical reactions in living systems. Being reactive, RNA is unstable.

Statement II: DNA evolved from RNA and is a more stable, genetic material. Its double helical strands being complementary, resist changes by evolving repairing mechanism.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (a) Statement I is correct but statement II is incorrect.
(b) Statement I is incorrect but statement II is correct.
(c) Both statement I and statement II are correct.
(d) Both statement I and statement II are incorrect.

125. Given below are two statements:

Statement I: Transfer RNAs and ribosomal RNA do not interact with mRNA.

Statement II: RNA interference (RNAi) takes place in all eukaryotic organisms as a method of cellular defence.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (a) Statement I is correct but statement II is incorrect.
(b) Statement I is incorrect but statement II is correct.
(c) Both statement I and statement II are correct.
(d) Both statement I and statement II are incorrect.

126. Match List-I with List-II.

List-I	List-II
A. Heart	I. Erythropoietin
B. Kidney	II. Aldosterone
C. Gastro-intestinal tract	III. Atrial natriuretic factor
D. Adrenal cortex	IV. Secretin

Choose the correct answer from the options given below:

- (a) A-I, B-III, C-IV, D-II (b) A-III, B-I, C-IV, D-II
(c) A-II, B-I, C-III, D-IV (d) A-IV, B-III, C-II, D-I

127. All living members of the Class Cyclostomata are

- (a) symbiotic (b) ectoparasite
(c) free-living (d) endoparasite.

128. Streptokinase produced by bacterium *Streptococcus* is used for

- (a) liver disease treatment
(b) removing clots from blood vessels
(c) curd production
(d) ethanol production.

129. Role of the water vascular system in Echinoderms is:

- A. Respiration and Locomotion
B. Excretion and Locomotion
C. Capture and transport of food
D. Digestion and Respiration
E. Digestion and Excretion

Choose the correct answer from the options given below:

- (a) B and C only (b) B, D and E only
(c) A and B only (d) A and C only.

130. Match List-I with List-II.

List-I	List-II
A. Pteridophyte	I. <i>Salvia</i>
B. Bryophyte	II. <i>Ginkgo</i>
C. Angiosperm	III. <i>Polytrichum</i>
D. Gymnosperm	IV. <i>Salvinia</i>

Choose the option with all correct matches.

- (a) A-III, B-IV, C-I, D-II (b) A-IV, B-III, C-II, D-I
(c) A-III, B-IV, C-II, D-I (d) A-IV, B-III, C-I, D-II

131. Which are correct?

- A. Computed tomography and magnetic resonance imaging detect cancers of internal organs.
B. Chemotherapeutics drugs are used to kill non-cancerous cells.
C. α -interferon activate the cancer patients' immune system and helps in destroying the tumour.
D. Chemotherapeutic drugs are biological response modifiers.
E. In the case of leukaemia blood cell counts are decreased.

Choose the correct answer from the options given below.

- (a) C and D only (b) A and C only
(c) B and D only (d) D and E only

132. What are the potential drawbacks in adoption of the IVF method?

- A. High fatality risk to mother
B. Expensive instruments and reagents
C. Husband/wife necessary for being donors
D. Less adoption of orphans
E. Not available in India
F. Possibility that the early embryo does not survive

Choose the correct answer from the options given below.

- (a) A, B, C, D only (b) A, B, C, E, F only
(c) B, D, F only (d) A, C, D, F only

133. Consider the following:

- A. The reductive division for the human female gametogenesis starts earlier than that of the male gametogenesis.
B. The gap between the first meiotic division and the second meiotic division is much shorter for males compared to females.
C. The first polar body is associated with the formation of the primary oocyte.
D. Luteinizing Hormone (LH) surge leads to disintegration of the endometrium and onset of menstrual bleeding.

Choose the correct answer from the options given below:

- (a) B and D are true. (b) B and C are true.
(c) A and B are true. (d) A and C are true.

134. In bryophytes, the gemmae help in which one of the following?

- (a) Nutrient absorption (b) Gaseous exchange
(c) Sexual reproduction (d) Asexual reproduction

135. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The primary function of the Golgi apparatus is to package the materials made by the endoplasmic reticulum and deliver it to intracellular targets and outside the cell.

Reason (R): Vesicles containing materials made by the endoplasmic reticulum fuse with the cis face of the Golgi apparatus, and they are modified and released from the trans face of the Golgi apparatus.

In the light of the above statements, choose the correct answer from the options given below:

- (a) A is true but R is false.
(b) A is false but R is true.
(c) Both A and R are true and R is the correct explanation of A.
(d) Both A and R are true but R is not the correct explanation of A.

136. Which one of the following statements refers to Reductionist Biology?

- (a) Chemical approach to study and understand living organisms.
(b) Behavioural approach to study and understand living organisms.
(c) Physico-chemical approach to study and understand living organisms.
(d) Physiological approach to study and understand living organisms.

137. After maturation, in primary lymphoid organs, the lymphocytes migrate for interaction with antigens to secondary lymphoid organ(s)/tissue(s) like

- A. thymus B. bone marrow
C. spleen D. lymph nodes
E. Peyer's patches

Choose the correct answer from the options given below:

- (a) E, A, B only (b) C, D, E only
(c) B, C, D only (d) A, B, C only.

138. Match List I with List II.

List I	List II
A. The Evil Quartet	I. Cryopreservation
B. <i>Ex situ</i> conservation	II. Alien species invasion
C. <i>Lantana camara</i>	III. Causes of biodiversity losses
D. Dodo	IV. Extinction

Choose the option with all correct matches.

- (a) A-III, B-IV, C-II, D-I (b) A-III, B-II, C-IV, D-I
(c) A-III, B-II, C-I, D-IV (d) A-III, B-I, C-II, D-IV

139. How many meiotic and mitotic divisions need to occur for the development of a mature female gametophyte from the megaspore mother cell in an angiosperm plant?

- (a) 1 Meiosis and 3 Mitosis
(b) No Meiosis and 2 Mitosis
(c) 2 Meiosis and 3 Mitosis
(d) 1 Meiosis and 2 Mitosis

140. Which of the following type of immunity is present at the time of birth and is a non-specific type of defence in the human body?

- (a) Cell-mediated Immunity
(b) Humoral Immunity
(c) Acquired Immunity
(d) Innate Immunity

141. Given below are two statements:

Statement I: Fig fruit is a non-vegetarian fruit as it has enclosed fig wasps in it.

Statement II: Fig wasp and fig tree exhibit mutual relationship as fig wasp completes its life cycle in fig fruit and fig fruit gets pollinated by fig wasp.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (a) Statement I is correct but statement II is incorrect.
(b) Statement I is incorrect but statement II is correct.
(c) Both statement I and statement II are correct.
(d) Both statement I and statement II are incorrect.

142. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Cells of the tapetum possess dense cytoplasm and generally have more than one nucleus.

Reason (R): Presence of more than one nucleus in the tapetum increases the efficiency of nourishing the developing microspore mother cells.

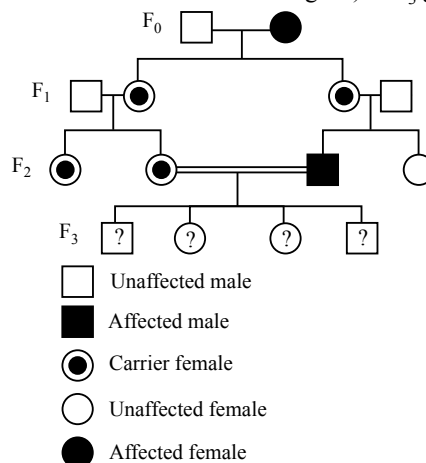
In light of the above statements, choose the most appropriate answer from the options given below:

- (a) A is true but R is false.
(b) A is false but R is true.
(c) Both A and R are true and R is the correct explanation of A.
(d) Both A and R are true but R is not the correct explanation of A.

143. From the statements given below choose the correct option.

- A. The eukaryotic ribosomes are 80S and prokaryotic ribosomes are 70S.
B. Each ribosome has two sub-units.
C. The two sub-units of 80S ribosome are 60S and 40S while that of 70S are 50S and 30S.
D. The two sub-units of 80S ribosome are 60S and 20S and that of 70S are 50S and 20S.

- E. The two sub-units of 80S are 60S and 30S and that of 70S are 50S and 30S.
 (a) A, B, E are true. (b) B, D, E are true.
 (c) A, B, C are true. (d) A, B, D are true.
144. Which one of the following enzymes contains 'Haem' as the prosthetic group?
 (a) Succinate dehydrogenase
 (b) Catalase
 (c) RuBisCO
 (d) Carbonic anhydrase
145. What is the name of the blood vessel that carries deoxygenated blood from the body to the heart in a frog?
 (a) Pulmonary vein (b) Vena cava
 (c) Aorta (d) Pulmonary artery
146. Given below are the stages in the life cycle of pteridophytes. Arrange the following stages in the correct sequence.
 A. Prothallus stage
 B. Meiosis in spore mother cells
 C. Fertilisation
 D. Formation of archegonia and antheridia in gametophyte.
 E. Transfer of antherozoids to the archegonia in presence of water.
 Choose the correct answer from the options given below:
 (a) D, E, C, A, B (b) E, D, C, B, A
 (c) B, A, D, E, C (d) B, A, E, C, D.
147. The blue and white selectable markers have been developed which differentiate recombinant colonies from non-recombinant colonies on the basis of their ability to produce colour in the presence of a chromogenic substrate. Given below are two statements about this method:
Statement I: The blue coloured colonies have DNA insert in the plasmid and they are identified as recombinant colonies.
Statement II: The colonies without blue colour have DNA insert in the plasmid and are identified as recombinant colonies.
 In the light of the above statements, choose the most appropriate answer from the options given below:
 (a) Statement I is correct but statement II is incorrect.
 (b) Statement I is incorrect but statement II is correct.
 (c) Both Statement I and statement II are correct.
 (d) Both Statement I and statement II are incorrect.
148. Which of the following microbes is not involved in the preparation of household products?
 A. *Aspergillus niger*
 B. *Lactobacillus*
 C. *Trichoderma polysporum*
 D. *Saccharomyces cerevisiae*
 E. *Propionibacterium shermanii*
 Choose the correct answer from the options given below:
 (a) C and D only (b) C and E only
 (c) A and B only (d) A and C only.
149. Silencing of specific mRNA is possible via RNAi because of
 (a) Complementary tRNA
 (b) Non-complementary ssRNA
 (c) Complementary dsRNA
 (d) Inhibitory ssRNA.
150. The complex II of mitochondrial electron transport chain is also known as
 (a) Cytochrome *c* oxidase
 (b) NADH dehydrogenase
 (c) Cytochrome *bc*₁
 (d) Succinate dehydrogenase.
151. While trying to find out the characteristic of a newly found animal, a researcher did the histology of adult animal and observed a cavity with presence of mesodermal tissue towards the body wall but no mesodermal tissue was observed towards the alimentary canal. What could be the possible coelom of that animal?
 (a) Schizocoelomate (b) Spongocoelomate
 (c) Acoelomate (d) Pseudocoelomate
152. Given below are two statements:
Statement I: In a floral formula $\oplus$ stands for zygomorphic nature of the flower, and $\underline{G}$ stands for inferior ovary.
Statement II: In a floral formula $\oplus$ stands for actinomorphic nature of the flower and $\underline{G}$ stands for superior ovary.
 In the light of the above statements, choose the correct answer from the options given below:
 (a) Statement I is correct but statement II is incorrect.
 (b) Statement I is incorrect but statement II is correct.
 (c) Both statement I and statement II are correct.
 (d) Both statement I and statement II are incorrect.
153. Given below are two statements:
Statement I: In ecosystem, there is unidirectional flow of energy of sun from producers to consumers.
Statement II: Ecosystems are exempted from 2nd law of thermodynamics.
 In the light of the above statements, choose the most appropriate answer from the options given below:
 (a) Statement I is correct but statement II is incorrect.
 (b) Statement I is incorrect but statement II is correct.
 (c) Both statement I and statement II are correct.
 (d) Both statement I and statement II are incorrect.
154. Which of the following is the unit of productivity of an ecosystem?
 (a) KCal m^{-3} (b) $(\text{KCal m}^{-2})\text{yr}^{-1}$
 (c) gm^{-2} (d) KCal m^{-2}
155. With the help of given pedigree, find out the probability for the birth of a child having no disease and being a carrier (has the disease mutation in one allele of the gene) in F₃ generation.



- (a) 1/8 (b) Zero (c) 1/4 (d) 1/2

156. In the seeds of cereals the outer covering of endosperm separates the embryo by a protein-rich layer called
 (a) integument (b) aleurone layer
 (c) coleoptile (d) coleorhiza.

157. Match List I with List II.

List I

- A. Chlorophyll *a*
 B. Chlorophyll *b*
 C. Xanthophylls
 D. Carotenoids

List II

- I. Yellow-green
 II. Yellow
 III. Blue-green
 IV. Yellow to Yellow-orange

Choose the option with all correct matches.

- (a) A-I, B-II, C-IV, D-III (b) A-I, B-IV, C-III, D-II
 (c) A-III, B-IV, C-II, D-I (d) A-III, B-I, C-II, D-IV

158. Who proposed that the genetic code for amino acids should be made up of three nucleotides?

- (a) Jacques Monod (b) Franklin Stahl
 (c) George Gamow (d) Francis Crick

159. Histones are enriched with

- (a) phenylalanine and leucine
 (b) phenylalanine and arginine
 (c) lysine and arginine
 (d) leucine and lysine.

160. Which of the following enzyme(s) are not essential for gene cloning?

- A. Restriction enzymes
 B. DNA ligase
 C. DNA mutase
 D. DNA recombinase
 E. DNA polymerase

Choose the correct answer from the options given below.

- (a) D and E only (b) B and C only
 (c) C and D only (d) A and B only

161. A specialised membranous structure in a prokaryotic cell which helps in cell wall formation, DNA replication and respiration is

- (a) cristae (b) endoplasmic reticulum
 (c) mesosome (d) chromatophores.

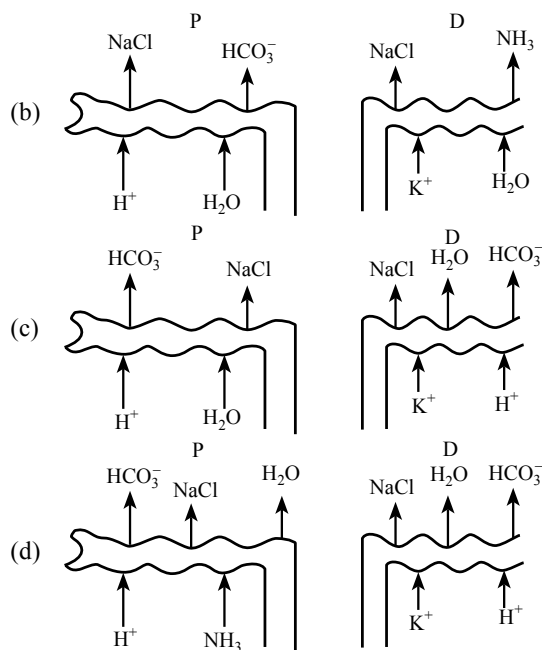
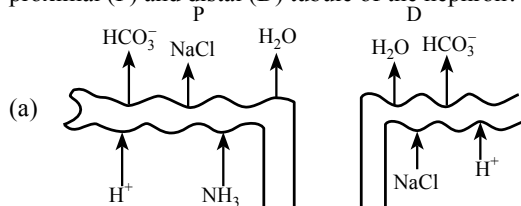
162. Which factor is important for termination of transcription?

- (a) ρ (rho) (b) γ (gamma)
 (c) α (alpha) (d) σ (sigma)

163. Which of the following statement is correct about location of the male frog copulatory pad?

- (a) Second digit of fore limb
 (b) First digit of the fore limb
 (c) First and second digit of fore limb
 (d) First digit of hind limb

164. Which of the following diagrams is correct with regard to the proximal (P) and distal (D) tubule of the nephron?



165. Identify the statement that is not correct.

- (a) Antigen binding site is located at C-terminal region of antibody molecules.
 (b) Constant region of heavy and light chains are located at C-terminus of antibody molecules.
 (c) Each antibody has two light and two heavy chains.
 (d) The heavy and light chains are held together by disulfide bonds.

166. Match List I with List II.

List I

- (A) Scutellum
 (B) Non-albuminous seed
 (C) Epiblast
 (D) Perisperm

List II

- I. Persistent nucellus
 II. Cotyledon of monocot seed
 III. Groundnut
 IV. Rudimentary cotyledon

Choose the option with all correct matches.

- (a) A-IV, B-III, C-I, D-II
 (b) A-II, B-IV, C-III, D-I
 (c) A-II, B-III, C-IV, D-I
 (d) A-IV, B-III, C-II, D-I

167. Find the statement that is not correct with regard to the structure of monocot stem.

- (a) Vascular bundles are conjoint and closed.
 (b) Phloem parenchyma is absent.
 (c) Hypodermis is parenchymatous.
 (d) Vascular bundles are scattered.

168. Twins are born to a family that lives next door to you. The twins are a boy and a girl. Which of the following must be true?

- (a) They were conceived through *in vitro* fertilisation.
 (b) They have 75% identical genetic content.
 (c) They are monozygotic twins.
 (d) They are fraternal twins.

169. Sweet potato and potato represent a certain type of evolution. Select the correct combination of terms to explain the evolution.

- (a) Homology, Convergent (b) Analogy, Divergent
(c) Analogy, Convergent (d) Homology, Divergent

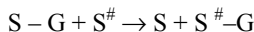
170. Which one of the following phytohormones promotes nutrient mobilization which helps in the delay of leaf senescence in plants?

- (a) Gibberellin (b) Cytokinin
(c) Ethylene (d) Absciscic acid

171. Why can't insulin be given orally to diabetic patients?

- (a) Because of structural variation.
(b) Its bioavailability will be increased.
(c) Human body will elicit strong immune response.
(d) It will be digested in Gastro-Intestinal (GI) tract.

172. Name the class of enzyme that usually catalyse the following reaction :



Where, G $\rightarrow$ a group other than hydrogen

S $\rightarrow$ a substrate

S[#] $\rightarrow$ another substrate

- (a) Transferase (b) Ligase
(c) Hydrolase (d) Lyase

173. Given below are two statements :

Statement I : The DNA fragments extracted from gel electrophoresis can be used in construction of recombinant DNA.

Statement II : Smaller size DNA fragments are observed near anode while larger fragments are found near the wells in an agarose gel.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (a) Statement I is correct but statement II is incorrect.
(b) Statement I is incorrect but statement II is correct.
(c) Both statement I and statement II are correct.
(d) Both statement I and statement II are incorrect.

174. The correct sequence of events in the life cycle of bryophytes is

- A. Fusion of antherozoid with egg.
B. Attachment of gametophyte to substratum.
C. Reduction division to produce haploid spores.
D. Formation of sporophyte.
E. Release of antherozoids into water.

Choose the correct answer from the options given below:

- (a) B, E, A, D, C (b) D, E, A, B, C
(c) D, E, A, C, B (d) B, E, A, C, D

175. Genes R and Y follow independent assortment. If RRRYY produce round yellow seeds and rrryy produce wrinkled green seeds, what will be the phenotypic ratio of the F₂ generation?

- (a) Phenotypic ratio – 9 : 3 : 3 : 1
(b) Phenotypic ratio – 9 : 7
(c) Phenotypic ratio – 1 : 2 : 1
(d) Phenotypic ratio – 3 : 1

176. Each of the following characteristics represent a Kingdom proposed by Whittaker. Arrange the following in increasing order of complexity of body organization.

- A. Multicellular heterotrophs with cell wall made of chitin.
B. Heterotrophs with tissue/organ/organ system level of body organization.
C. Prokaryotes with cell wall made of polysaccharides and amino acids.
D. Eukaryotic autotrophs with tissue/organ level of body organization.
E. Eukaryotes with cellular body organization.

Choose the correct answer from the options given below:

- (a) A, C, E, D, B (b) C, E, A, B, D
(c) A, C, E, B, D (d) C, E, A, D, B

177. Match List-I with List-II.

List I

- A. Centromere
B. Cilium
C. Cristae
D. Cell membrane

List II

- I. Mitochondrion
II. Cell division
III. Cell movement
IV. Phospholipid Bilayer

Choose the correct answer from the options given below:

- (a) A-IV, B-II, C-III, D-I (b) A-II, B-III, C-I, D-IV
(c) A-I, B-II, C-III, D-IV (d) A-II, B-I, C-IV, D-III

178. Which one of the following equations represents the Verhulst-Pearl Logistic Growth of population?

- (a) $\frac{dN}{dt} = rN \left(\frac{N-K}{N} \right)$ (b) $\frac{dN}{dt} = N \left(\frac{r-K}{K} \right)$
(c) $\frac{dN}{dt} = r \left(\frac{K-N}{K} \right)$ (d) $\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$

179. Match List-I with List-II.

List-I

- A. Emphysema
B. Angina Pectoris
C. Glomerulonephritis
D. Tetany

List-II

- I. Rapid spasms in muscle due to low Ca⁺⁺ in body fluid
II. Damaged alveolar walls and decreased respiratory surface
III. Acute chest pain when not enough oxygen is reaching to heart muscle
IV. Inflammation of glomeruli of kidney

Choose the correct answer from the options given below :

- (a) A-II, B-IV, C-III, D-I (b) A-II, B-III, C-IV, D-I
(c) A-III, B-I, C-IV, D-II (d) A-III, B-I, C-II, D-IV

180. Cardiac activities of the heart are regulated by:

- A. Nodal tissue
B. A special neural centre in the medulla oblongata
C. Adrenal medullary hormones
D. Adrenal cortical hormones

Choose the correct answer from the options given below:

- (a) A, C and D only (b) A, B and D only
(c) A, B and C only (d) A, B, C and D

mtg

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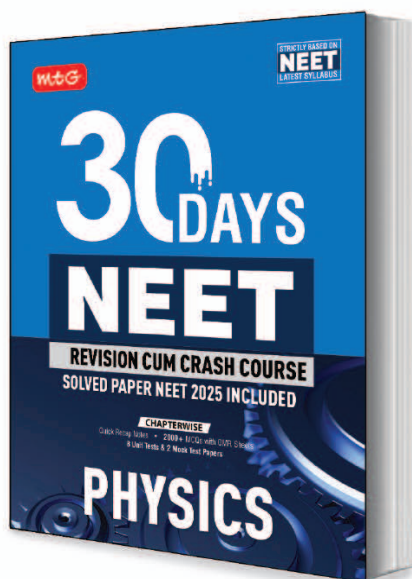
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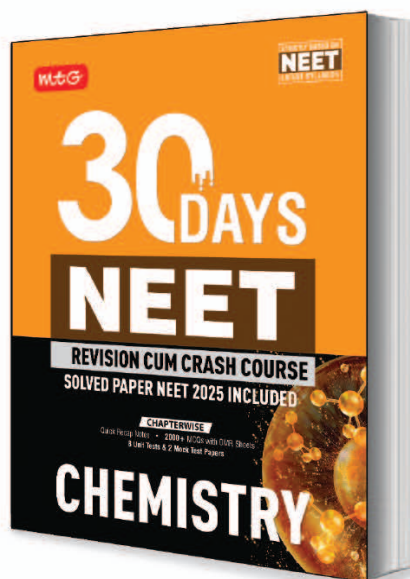
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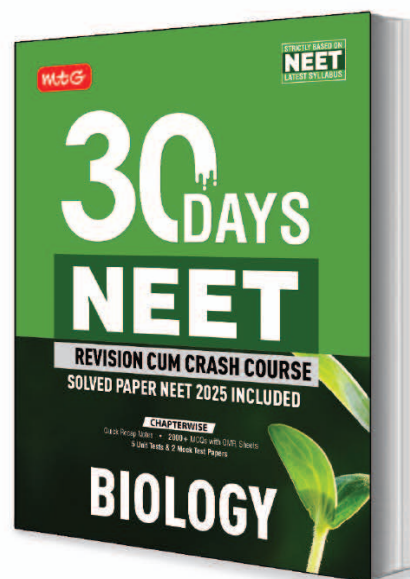
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ANSWER KEY**PHYSICS**

1. (a) 2. (d) 3. (c) 4. (a) 5. (d) 6. (d) 7. (a) 8. (d) 9. (b) 10. (a)
 11. (d) 12. (d) 13. (d) 14. (b) 15. (d) 16. (d) 17. (b) 18. (b) 19. (d) 20. (d)
 21. (d) 22. (c) 23. (d) 24. (b) 25. (b) 26. (a) 27. (a) 28. (d) 29. (a) 30. (b)
 31. (a) 32. (b) 33. (c) 34. (a) 35. (a) 36. (a) 37. (d) 38. (c) 39. (d) 40. (c)
 41. (d) 42. (c) 43. (a) 44. (c) 45. (d)

CHEMISTRY

46. (d) 47. (a) 48. (b) 49. (c) 50. (d) 51. (c,d) 52. (a) 53. (c) 54. (c) 55. (a)
 56. (a) 57. (a) 58. (b) 59. (c) 60. (b) 61. (d) 62. (c) 63. (d) 64. (c) 65. (d)
 66. (a) 67. (a) 68. (b) 69. (a) 70. (d) 71. (d) 72. (a) 73. (b) 74. (c) 75. (c)
 76. (b) 77. (c) 78. (d) 79. (a) 80. (d) 81. (c) 82. (c) 83. (c) 84. (d) 85. (b)
 86. (b) 87. (d) 88. (b) 89. (c) 90. (d)

BIOLOGY

91. (a) 92. (a) 93. (a) 94. (a) 95. (d) 96. (d) 97. (c) 98. (a) 99. (b) 100. (d)
 101. (c) 102. (c) 103. (c) 104. (a) 105. (a) 106. (b) 107. (d) 108. (d) 109. (d) 110. (c)
 111. (b) 112. (b) 113. (a) 114. (d) 115. (c) 116. (d) 117. (a) 118. (d) 119. (c) 120. (a)
 121. (d) 122. (a) 123. (d) 124. (c) 125. (b) 126. (b) 127. (b) 128. (b) 129. (d) 130. (d)
 131. (b) 132. (c) 133. (c) 134. (d) 135. (c) 136. (c) 137. (b) 138. (d) 139. (a) 140. (d)
 141. (d) 142. (c) 143. (c) 144. (b) 145. (b) 146. (c) 147. (b) 148. (d) 149. (c) 150. (d)
 151. (d) 152. (b) 153. (a) 154. (b) 155. (c) 156. (b) 157. (d) 158. (c) 159. (c) 160. (c)
 161. (c) 162. (a) 163. (b) 164. (d) 165. (a) 166. (c) 167. (c) 168. (d) 169. (c) 170. (b)
 171. (d) 172. (a) 173. (c) 174. (a) 175. (a) 176. (d) 177. (b) 178. (d) 179. (b) 180. (d)

Explanations

1. (a) : The magnetic field arising due to displacement current is non zero everywhere with maximum at the imaginary cylindrical surface connecting peripheries of the plates. This is because the displacement current is strongest there, where the changing electric field is most concentrated.

2. (d) : Change in potential energy in electric field,

$$\begin{aligned}\Delta U &= pE(\cos\theta_1 - \cos\theta_2) \\ &= pE[(\cos(0^\circ) - \cos(60^\circ))] \\ \therefore \Delta U &= \frac{pE}{2} = \frac{5 \times 10^{-6} \times 4 \times 10^5}{2} \\ \Delta U &= 1 \text{ J}\end{aligned}$$

3. (c) : Mass of ball, $m = 0.5 \text{ kg}$

$$h_1 = 40 \text{ m and } h_2 = 10 \text{ m}$$

So, velocity of ball before hitting the ground

$$v_1 = \sqrt{2gh_1} = \sqrt{2 \times 9.8 \times 40}$$

$$v_1 = 28 \text{ m/s}$$

After hitting the ground, let final velocity of the ball is v_2 .

$$\therefore v_2 = \sqrt{2gh_2} = \sqrt{2 \times 9.8 \times 10}$$

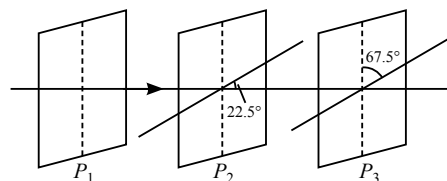
$$v_2 = 14 \text{ m/s}$$

Change in momentum, $\Delta p = m[v_2 - (-v_1)]$

$$= \frac{1}{2}[14 - (-28)]$$

$$\Delta p = 21 \text{ kg m/s} \Rightarrow \text{Impulse, } \Delta p = 21 \text{ N s}$$

4. (a) :



According to the Malus law,

$$I = I_0 \cos^2(\theta_1)$$

$$I_1 = I_0$$

$$I_2 = I_1 \cos^2(\theta_2)$$

$$I_2 = I_0 \cos^2(22.5)$$

$$I_2 = I_0 (0.85)$$

$$I_3 = I_2 \cos^2(\theta_3) = I_0 \times 0.85 \times \cos^2(67.5^\circ)$$

$$I_3 = I_0 \times 0.125$$

$$\text{or } I_3 = I_0/8$$

5. (d) : According to work-energy theorem,

Work done = change in kinetic energy

$$W = \Delta KE$$

For car A,

$$W_A = (KE)_A$$

$$F_A S_A = (KE)_A \quad \dots(i)$$

Similarly for car B,

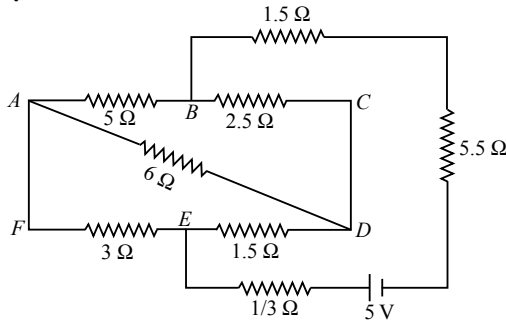
$$F_B S_B = (KE)_B \quad \dots(ii)$$

From eqn. (i) and (ii)

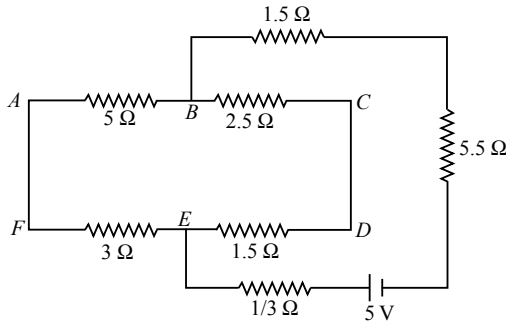
$$\frac{(KE)_A}{(KE)_B} = \frac{F_A S_A}{F_B S_B}$$

$$\frac{100}{225} = \frac{F_A}{F_B} \times \frac{1000}{1500} \Rightarrow \frac{F_A}{F_B} = \frac{2}{3}$$

6. (d) :



Circuit $ABCDEF$ form wheatstone bridge,
So, resistance across arm AD , $R_{AD} = 6 \Omega$ can be eliminated.



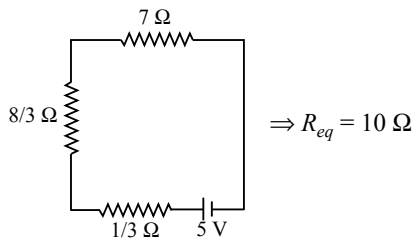
From circuit $ABCDEF$:

Resistance 5Ω and 3Ω are in series and 2.5Ω and 1.5Ω are in series. Then both the series combination are in parallel with each other. Thus equivalent resistance is

$$\frac{1}{R_{eq}} = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}$$

$$\therefore R_{eq} = \frac{8}{3} \Omega$$

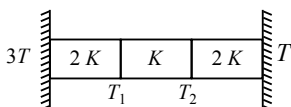
Now, the given circuit can be redrawn as



Current through battery,

$$I = \frac{5}{10} = 0.5 \text{ A}$$

7. (a) :



According to heat transfer, at junction T_1

$$\frac{2KA(3T - T_1)}{L} = \frac{KA(T_1 - T_2)}{L}$$

Since, all identical rods are connected in series,

So, A and L are same.

$$\therefore 6T - 2T_1 = T_1 - T_2$$

$$6T = 3T_1 - T_2 \quad \dots(i)$$

At junction T_2 ,

$$\frac{KA(T_1 - T_2)}{L} = \frac{2KA(T_2 - T)}{L}$$

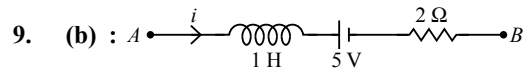
$$\text{or } T_1 - T_2 = 2T_2 - 2T$$

$$\text{or } 2T = 3T_2 - T_1 \quad \dots(ii)$$

From eqn. (i) and (ii)

$$\frac{T_1}{T_2} = \frac{5}{3}$$

8. (d) : As the box oscillates, sand leaks slowly out of the box which leads to decrease in effective mass of the bob. As, frequency is inversely proportional to the square root of m , hence frequency (ω) increases and amplitude will decrease due to dissipation of energy.



From the diagram (Using KVL)

$$V_A - L \frac{di}{dt} - 5 - 2i = V_B$$

Using, $i = 2 \text{ A}$, $\frac{di}{dt} = 1 \text{ A/sec}$ and $L = 1 \text{ H}$, we get

$$V_A - 1 - 5 - 4 = V_B$$

$$\Rightarrow V_A - V_B = 10 \text{ V}$$

10. (a) : As force (F) is constant, so

$$F = \frac{mv^2}{r}$$

$$\text{We can write, } F = \frac{m^2 v^2 r^2}{mr^3} = \frac{L^2}{mr^3} \quad (\text{As, } L = mvr)$$

As force is constant, so $L^2 \propto mr^3$

$$\text{Using, } L = \frac{nh}{2\pi}, \text{ we get } \frac{n^2 h^2}{4\pi^2} = mr^3$$

$$\Rightarrow n^2 \propto r^3 \quad \text{or} \quad r \propto n^{2/3}$$

$$\text{Also, } mvr = \frac{nh}{2\pi}$$

$$\text{As } r \propto n^{2/3}, \text{ so } mv(n^{2/3}) \propto \frac{nh}{2\pi}$$

$$\text{or } v \propto n^{1/3}$$

11. (d) : Given, $t = x^2 + x$

Differentiating with respect to ' t ' both sides,

$$\Rightarrow \frac{dt}{dt} = 2x \frac{dx}{dt} + \frac{dx}{dt}$$

$$\Rightarrow 1 = 2xv + v \quad \dots(i)$$

$$\Rightarrow 1 = v(2x + 1)$$

$$v = \frac{1}{2x + 1} \quad \dots(ii)$$

Again differentiating eq (i) with respect to ' t ',

$$0 = 2 \left(\frac{dx}{dt} \cdot v + x \frac{dv}{dt} \right) + \frac{dv}{dt}$$

$$0 = 2(v^2 + x.a) + a$$

$$0 = 2v^2 + 2xa + a$$

$$0 = [2v^2 + a(2x+1)]$$

$$\Rightarrow 2v^2 = -a(2x+1)$$

Using equation (ii), we get

$$2\left(\frac{1}{2x+1}\right)^2 = -a(2x+1)$$

$$\therefore a = \frac{-2}{(2x+1)^3}$$

12. (d) : Flux, $\phi = B \times \pi r^2$

$$\text{or } \frac{nh}{e} = B \times \pi r^2 \quad \text{or } r^2 = \frac{nh}{Be\pi} \quad \dots(i)$$

$$\text{Also, } Bev = \frac{mv^2}{r} \quad \text{or } mv = Ber \quad \dots(ii)$$

Let magnetic moment of electron in circular orbit of an atom is μ .

$$\begin{aligned} \mu &= IA \\ &= \frac{e}{T} A = \frac{e\omega}{2\pi} (\pi r^2) = \frac{evr}{2} \quad \dots(iii) \end{aligned}$$

where m is the mass of electron and L is angular momentum.

From eq. (iii),

$$\mu = \frac{evr}{2} = \frac{e}{2m} L \quad (\because L = mvr)$$

$$\text{or } \mu = \frac{e}{2m} \times mvr = \frac{e}{2m} \times Ber^2 \quad (\text{Using eq (ii)})$$

$$\text{or } \mu = \frac{e}{2m} \times Be \times \frac{nh}{Be\pi} \quad (\text{Using eq (i)})$$

$$\therefore \mu = \frac{he}{2\pi m}$$

13. (d) : Focal length of objective lens, $f_0 = 2$ cm

Focal length of eyepiece lens, $f_e = 4$ cm

Distance of distinct vision, $D = 25$ cm

$$m = \frac{L}{f_0} \times \frac{D}{f_e} = \frac{40}{2} \times \frac{25}{4}$$

$$m = 125$$

14. (b) : Given : Angle of inclination, $\theta = 45^\circ$

$$t_{\text{rough}} = 2t_{\text{smooth}}$$

Length of both inclined surfaces is equal.

For smooth surface, initial velocity is zero, i.e., $u = 0$, the distance 'S' is given by

$$S = \frac{1}{2} a_{\text{smooth}} t_{\text{smooth}}^2$$

Acceleration of smooth plane is $a_{\text{smooth}} = g \sin \theta$

Using $\theta = 45^\circ$, we get

$$a_{\text{smooth}} = g \sin 45^\circ = \frac{g}{\sqrt{2}} \quad \text{or } \frac{g\sqrt{2}}{2}$$

$$\Rightarrow S = \frac{1}{2} g \frac{\sqrt{2}}{2} t_{\text{smooth}}^2$$

$$\Rightarrow t_{\text{smooth}} = \sqrt{\frac{4S}{g\sqrt{2}}}$$

For rough surface :

$$S = \frac{1}{2} a_{\text{rough}} t_{\text{rough}}^2$$

$$(\because u = 0)$$

$$a_{\text{rough}} = g \sin \theta - \mu_k g \cos \theta$$

$$\text{Using } \theta = 45^\circ, \text{ we get } a_{\text{rough}} = \frac{g\sqrt{2}}{2} - \mu_k g \frac{\sqrt{2}}{2}$$

$$\therefore S = \frac{1}{2} \left(g \frac{\sqrt{2}}{2} - \mu_k g \frac{\sqrt{2}}{2} \right) t_{\text{rough}}^2$$

$$\text{or } t_{\text{rough}} = \sqrt{\frac{2S}{g \frac{\sqrt{2}}{2} (1 - \mu_k)}} = \sqrt{\frac{4S}{g\sqrt{2} (1 - \mu_k)}}$$

$$\text{As, } t_{\text{rough}} = 2t_{\text{smooth}}$$

$$\sqrt{\frac{4S}{g\sqrt{2} (1 - \mu_k)}} = 2 \sqrt{\frac{4S}{g\sqrt{2}}}$$

Squaring on both sides, we get

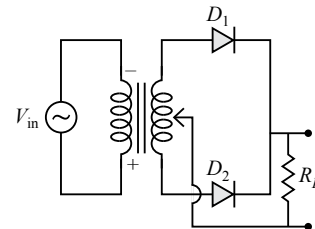
$$\Rightarrow \frac{4S}{g\sqrt{2} (1 - \mu_k)} = 4 \cdot \frac{4S}{g\sqrt{2}}$$

$$\Rightarrow 4 = \frac{1}{1 - \mu_k}$$

$$\Rightarrow 4(1 - \mu_k) = 1 \quad \text{or } 1 = 4 - 4\mu_k$$

$$\mu_k = \frac{3}{4} = 0.75$$

15. (d) : Given that, $V_{in} = 220 \sin(100\pi t)$ volt and $t = 15$ msec

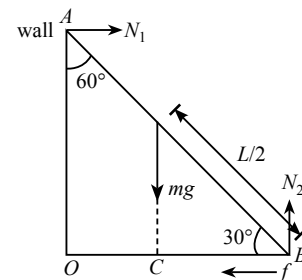


$$V_{in} = 220 \sin(100 \times \pi \times 15 \times 10^{-3})$$

$$= 220 \sin\left(\frac{3}{2}\pi\right) = -220 \text{ V}$$

Hence, D_1 is reverse biased and D_2 is forward biased.

16. (d) : Given that : Mass of rod, $m = 20$ kg, length of rod, $l = 5$ m



Weight of the uniform rod $= mg$

$$= 20 \times 10 = 200 \text{ N}$$

For vertical direction, $W = N_2$

For horizontal direction, $f = N_1$
(where f is the frictional force)

Since rod is stationary, so net torque is zero (about B)

$\Rightarrow$ Distance from wall to the centre of mass of rod ($L/2$) is

$$D = \frac{L}{2} \cos 30^\circ = \frac{1}{2} \left(\frac{\sqrt{3}}{2} \right) = \frac{\sqrt{3}}{4}$$

$\Rightarrow$ Now, distance $OA = L \sin 30^\circ = \frac{L}{2}$.

Taking torque about point B to be zero.

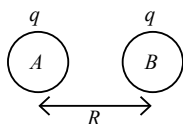
$$\Rightarrow mg \times \frac{L}{2} \cos 30^\circ = N_1 \times L \sin 30^\circ$$

$$\Rightarrow mg \times \frac{1}{2} \times \frac{\sqrt{3}}{2} = f \times \frac{1}{2} \quad [\text{As, } N_1 = f]$$

$$\Rightarrow 200 \times \frac{\sqrt{3}}{4} = \frac{f}{2} \text{ or } f = 100\sqrt{3} \text{ N}$$

17. (b) : According to Coulomb's law, $F = \frac{kq^2}{R^2}$

By touching an uncharged sphere C with charged sphere A , charge on each sphere A and C becomes $q/2$.



Now, when charged sphere C kept in contact with sphere B , then

$$\text{total charge is, } \frac{q}{2} + \frac{q}{1} = \frac{3q}{2}$$

Then,

$$\text{Charge on } B = \frac{3q}{4}$$

$$\text{Charge on } C = \frac{3q}{4}$$

Now, charge on A is $\frac{q}{2}$ and charge on B is $\frac{3q}{4}$.

$$\text{So, } F' = \frac{k}{R^2} \left(\frac{q}{2} \right) \left(\frac{3q}{4} \right) = k \frac{3}{8} \frac{q^2}{R^2} \text{ or } F' = \frac{3}{8} F$$

18. (b) : The relative velocity of the bus in the same direction of motion of scooty $= V_B - V_S = V_B - 60$

The relative velocity of the bus in the opposite direction of motion of scooty $= V_B + V_S = V_B + 60$

Now, distance travelled by bus in time (T) $= V_B T$

Now, setting up equations for the distances between the buses in both cases and solve for T and V_B .

$$\frac{V_B T}{V_B - 60} = 30 \quad \dots(i)$$

$$\frac{V_B T}{V_B + 60} = 10 \quad \dots(ii)$$

By solving equations (i) and (ii), we get

$$T = 15 \text{ min and } V_B = 120 \text{ km/h}$$

19. (d) : As, pressure at equilibrium is given by,

$$P_{eq} = \frac{P_1 V_1 + P_2 V_2}{V_1 + V_2}$$

$$P_{eq} = \frac{1 \times 2 + 2 \times 3}{2 + 3}$$

$V_1 = 2l$	$V_2 = 3l$
$n_1 = 5 \text{ moles}$	$n_2 = 4 \text{ moles}$
$P_1 = 1 \text{ atm}$	$P_2 = 2 \text{ atm}$

$$P_{eq} = \frac{8}{5} = 1.6 \text{ atm}$$

$$20. (d) : 2\pi r = n\lambda \Rightarrow \lambda = \frac{2\pi r}{n} = \frac{2\pi \times (0.052) \frac{n^2}{Z}}{n}$$

$$\therefore \lambda = 2\pi \times (0.052) \times 2 \approx 0.67 \text{ nm}$$

21. (d) : Given, $V = 220 \text{ V}$, $\nu = 50 \text{ Hz}$
 $R = 20 \Omega$, $X_L = 45 \Omega$ and $X_C = 25 \Omega$

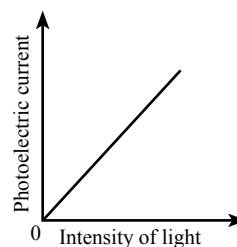
$$\text{Current, } I = \frac{V}{Z} = \frac{220}{\sqrt{R^2 + (X_L - X_C)^2}} = \frac{220}{\sqrt{400 + 400}}$$

$$= \frac{220}{20\sqrt{2}} = \frac{11}{\sqrt{2}} = 7.8 \text{ A}$$

$$\text{Phase angle, } \phi = \tan^{-1} \left(\frac{X_L - X_C}{R} \right) = \tan^{-1} \left(\frac{45 - 25}{20} \right)$$

$$= \tan^{-1} \left(\frac{20}{20} \right) \text{ or } \phi = 45^\circ$$

22. (c) : The graph between photoelectric current and intensity of light is given as



Photoelectric current is independent of frequency.

23. (d) : The fundamental frequency (f) is related to length of the pipe (L) and speed of sound in air (v) as

$$f = \frac{v}{\lambda}$$

For a pipe open at both ends, the fundamental frequency corresponding to a wavelength λ that is twice the length of the pipe is

$$f = \frac{v}{2L} \quad \dots(i)$$

When the pipe is dipped vertically in water so that half of its length is submerged, the new length of the air column becomes

$$L' = \frac{L}{2} \quad \dots(ii)$$

For the new length of the air column, the wavelength λ' is given by $\lambda' = 4L'$

$$\lambda' = 4 \times \frac{L}{2} \quad (\text{Using (ii)})$$

$$\lambda' = 2L$$

The new fundamental frequency,

$$f' = \frac{v}{\lambda'} = \frac{v}{2L} \quad \dots(iii)$$

Comparing eq. (i) and (iii), we get

$$f' = f$$

24. (b) : For a spring-mass system, the angular frequency is given by

$$\omega = \sqrt{\frac{k}{m}}$$

For body P : $v_{1,\max} = \omega_1 A_P = \sqrt{\frac{k_1}{m}} A_P$... (i)

For body Q : $v_{2,\max} = \omega_2 A_Q = \sqrt{\frac{k_2}{m}} A_Q$... (ii)

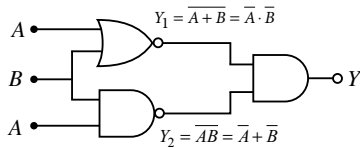
Equate the equation (i) and (ii),

$$\sqrt{\frac{k_1}{m}} A_P = \sqrt{\frac{k_2}{m}} A_Q \quad \text{or} \quad \sqrt{k_1} A_P = \sqrt{k_2} A_Q$$

Ratio of the amplitudes of Q and P

$$\frac{A_Q}{A_P} = \sqrt{\frac{k_1}{k_2}}$$

25. (b) :



$$\Rightarrow Y = (\overline{A \cdot B})(\overline{A + B})$$

$$= \overline{A \cdot B} + \overline{B \cdot A} = \overline{A \cdot B} = (A + B) = \text{NOR gate}$$

26. (a) : Initially volume of the oxygen cylinder,
 $V_1 = 30 \text{ L} = 30 \times 10^{-3} \text{ m}^3$

Number of moles, $n_1 = 18.20$

$$T_1 = 27 + 273 = 300 \text{ K}$$

Final gauge pressure, $P_g = 11 \text{ atm}$

Final absolute pressure, $P_2 = P_g + P_{\text{atm}} = 11 \text{ atm} + 1 \text{ atm}$
 $= 12 \text{ atm}$

For oxygen, molecular weight, $M = 32 \text{ g}$

Initial mass of oxygen in the cylinder,

$$m_1 = n_1 M = 18.20 \times 32 = 582.4 \text{ g}$$

Now, $V_2 = 30 \times 10^{-3} \text{ m}^3$, $P_2 = 12 \times 1.01 \times 10^5 \text{ Pa}$

$$n_2 = \frac{P_2 V_2}{RT_2} = \frac{(12 \times 1.01 \times 10^5) \times (30 \times 10^{-3})}{8.3 \times 300} = 14.60$$

$\therefore$ Final mass of oxygen gas in the cylinder,

$$m_2 = 14.60 \times 32 = 467.2 \text{ g}$$

$\therefore$ Mass of the withdrawn oxygen gas

$$= m_1 - m_2$$

$$= 582.4 - 467.2$$

$$= 115.2 \text{ kg} \approx 0.116 \text{ kg}$$

27. (a) : For thin lenses in contact, then the powers of each lens add up.

$$P_t = P_1 + P_2 + P_3 + P_4 = 4P$$

Given linear magnification of each lens m is

The overall magnification of combined lens system is

$$M = m \times m \times m \times m = m^4 \text{ (dimensionless).}$$

Hence the combination has power $4P$ and magnification m^4 .

$4P$ and m^4 – follows from the additive power law for co-axial lenses and the multiplicative rule for successive magnifications.

28. (d) : Given, pressure, $P = \text{constant} \Rightarrow P_A = P_B$

$$\text{Also, } \Delta U_A = \Delta U_B$$

From first law of thermodynamics

$$\Delta Q_A = \Delta U_A + \Delta W_A$$

$$\Rightarrow \Delta Q_A = \Delta U_A + P_A (16) \times A_A \quad \dots (i)$$

$$\Delta Q_B = \Delta U_B + \Delta W_B$$

$$\Rightarrow \Delta Q_B = \Delta U_B + P_B (9) \times A_B \quad \dots (ii)$$

Given $\Delta Q_A = \Delta Q_B$

$$\Rightarrow \Delta U_A + P_A (16) A_A = \Delta U_B + P_B (9) \times A_B$$

$$\Rightarrow 16 \times \pi \times r_A^2 = 9 \times \pi \times r_B^2$$

$$\Rightarrow \left(\frac{r_A}{r_B}\right)^2 = \frac{9}{16} \Rightarrow \frac{r_A}{r_B} = \frac{3}{4}$$

29. (a) : As, $T \propto S^\alpha A^\beta \rho^\gamma R^\delta$

$$[T] = [MT^{-2}]^\alpha [L^2]^\beta [ML^{-3}]^\gamma [L]^\delta$$

$$\alpha + \gamma = 0, 2\beta - 3\gamma + \delta = 0, -2\alpha = 1$$

$$\therefore \alpha = -1/2$$

$$\gamma = 1/2$$

$$\text{Now, } 2\beta - \frac{3}{2} + \delta = 0$$

$$2\beta + \delta = \frac{3}{2}$$

By checking with options

$$\beta = -1, \delta = \frac{7}{2}$$

$$\text{So, } v \propto \sqrt{\Delta P} \propto \sqrt{\frac{s}{r}} \propto r^{-\frac{1}{2}}$$

$$\text{So, } a = -\frac{1}{2}$$

30. (b) : Let speed at P is v .

$$PM = l \sin \theta$$

$$h = l + l \sin \theta$$

Using conservation of energy,

$$\frac{1}{2} m v_0^2 + 0 = mgh + \frac{1}{2} m v^2$$

$$\frac{1}{2} v_0^2 = gl(1 + \sin \theta) + \frac{1}{2} v^2$$

At P , $T = 0$

$$\text{So, } T - mg \cos(90^\circ + \theta) = \frac{mv^2}{l}$$

$$mg \sin \theta = \frac{mv^2}{l}$$

$$\text{or } \frac{v^2}{\sin \theta} = gl$$

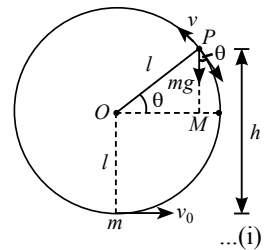
Putting in (i),

$$\frac{1}{2} v_0^2 = \frac{2 \cdot v^2}{2 \cdot \sin \theta} (1 + \sin \theta) + \frac{1}{2} v^2$$

$$\frac{1}{2} v_0^2 = \frac{v^2}{2} \left[\frac{2}{\sin \theta} (1 + \sin \theta) + 1 \right]$$

$$v_0^2 = v^2 \left[\frac{2 + 2 \sin \theta + \sin \theta}{\sin \theta} \right]$$

$$v_0^2 = v^2 \times \left[\frac{2 + 3 \sin \theta}{\sin \theta} \right]$$



$$\frac{v^2}{v_0^2} = \frac{\sin \theta}{2 + 3 \sin \theta}$$

$$\frac{v}{v_0} = \sqrt{\frac{\sin \theta}{2 + 3 \sin \theta}}$$

31. (a) : Given, $P = \frac{a^3 b^2}{c \sqrt{d}}$

Percentage error in quantity P ,

$$\frac{\Delta P}{P} \times 100 = \left(\frac{3\Delta a}{a} + \frac{2\Delta b}{b} + \frac{\Delta c}{c} + \frac{1}{2} \frac{\Delta d}{d} \right) \times 100$$

$$= 3(1) + 2(3) + 2 + \frac{1}{2} \times 4$$

$$= 3 + 6 + 2 + 2 = 13\%$$

32. (b) : $T_1 = 27$ days, $R_1 = R$, $R_2 = 2R$

By conservation of angular momentum

$$I_1 \omega_1 = I_2 \omega_2$$

$$\frac{2}{5} m R_1^2 \times \frac{2\pi}{T_1} = \frac{2}{5} m R_2^2 \times \frac{2\pi}{T_2}$$

$$\frac{R^2}{27} = \frac{4R^2}{T_2}; T_2 = 27 \times 4 = 108 \text{ days}$$

33. (c) : Given, $R_{\text{martian}} = 4R_{\text{mercury}}$

$$T_{\text{martian}} = 687 \text{ Earth year}$$

$$T_{\text{mercury}} = ?$$

By Kepler's third law

$$\left(\frac{T_{\text{Mercury}}}{T_{\text{Martian}}} \right)^2 = \left(\frac{R_{\text{Mercury}}}{R_{\text{Martian}}} \right)^3 = \left(\frac{1}{4} \right)^3 = \frac{1}{64}$$

$$T_{\text{Mercury}} = \frac{T_{\text{Martian}}}{8} = \frac{687}{8} = 85.8 \text{ days}$$

Nearest answer is 88 earth days.

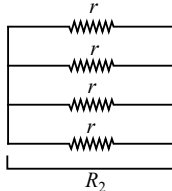
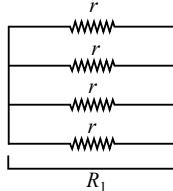
34. (a) : Resistance of each part, $r = R/8$

$$R_1 = R_2 = \frac{r}{4} = \frac{R}{32}$$

R_1 and R_2 are in series

$$\therefore R_{eq} = R_1 + R_2$$

$$= \frac{R}{32} + \frac{R}{32} = \frac{R}{16}$$



35. (a) : $\lambda_{ph} = \frac{hc}{E}$... (i)

$$\lambda_e = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$$
 ... (ii)

From eqn. (i) and (ii), we get

$$\therefore \frac{\lambda_{ph}}{\lambda_e} = \frac{hc\sqrt{2mE}}{E \cdot h} = c\sqrt{\frac{2m}{E}}$$

36. (a) : Moment of inertia of a solid sphere of mass M and radius R is

$$I = \frac{2}{5} MR^2$$

Mass of smaller sphere, $M_s = \frac{M}{8}$

Mass of remaining part, $M_r = M - \frac{M}{8} = \frac{7M}{8}$

Using parallel-axes theorem, moment of inertia of smaller sphere about y -axis,

$$I_s = \frac{2}{5} M_s R^2 + M_s R^2 = \left(\frac{2}{5} + 1 \right) M_s R^2 = \frac{7}{5} M_s R^2$$

or $I_s = \frac{7}{5} \times \frac{M}{8} \times R^2 = \frac{7}{40} MR^2$

Moment of inertia of the larger part of the sphere about y -axis,

$$I_L = \frac{2}{5} M (2R)^2 = \frac{8}{5} MR^2$$

Now, moment of inertia of the rest part of the sphere is

$$I_r = I_L - I_s = \frac{8}{5} MR^2 - \frac{7}{40} MR^2 = \frac{57}{40} MR^2$$

$$\therefore \frac{I_s}{I_r} = \frac{\frac{7}{40} MR^2}{\frac{57}{40} MR^2} = \frac{7}{57}$$

37. (d) : For an electron to move undeflected through perpendicular electric and magnetic fields, the net force on it must be zero.

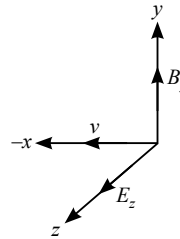
i.e., $F_B = F_E$

or $qvB = qE$

or $E = vB = \frac{c}{100} \times 9 \times 10^{-4}$

$$= \frac{3 \times 10^8}{100} \times 9 \times 10^{-4} = 27 \times 10^2 \text{ V/m}$$

38. (c) :



Given, $E_z = 60 \cos(5x + 1.5 \times 10^9 t) \text{ V/m}$

As, $B_0 = \frac{E_0}{c} \Rightarrow B_0 = \frac{60}{3 \times 10^8} = 2 \times 10^{-7} \text{ T}$

Since, In an em wave, the direction of wave propagation is along $\vec{E} \times \vec{B}$ thus magnetic field will be along y -direction.

$$\therefore B_y = 2 \times 10^{-7} \cos(5x + 1.5 \times 10^9 t) \text{ J}$$

39. (d) : Given, $W = 48 \text{ N}$ and $h = \frac{R}{3}$

Gravitational force at a height h above the surface of the Earth,

$$F = \frac{GmM}{(R+h)^2}$$

$$\text{or } F = \frac{G \left(\frac{W}{g} \right) \times M}{\left(R + \frac{R}{3} \right)^2} \quad \left[\because \text{Weight, } W = mg \text{ or } m = \frac{W}{g} \right]$$

$$\text{or } F = \frac{G \times 48 \times M \times 9}{g \times 16R^2}$$

Substituting, $g = \frac{GM}{R^2}$, we get

$$F = \frac{48 \times 9}{16} = 27 \text{ N}$$

40. (c) : Brewster's angle, $\tan \theta_B = \frac{\mu_2}{\mu_1}$

where μ_1 is the refractive index of air and μ_2 is the refractive index of the given medium, $\mu_2 = 1.73$.

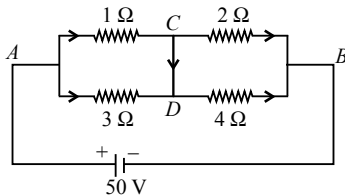
$$\therefore \theta_B = \tan^{-1}(1.73) \approx 60^\circ$$

By law of reflection,

$$i_p = r = 60^\circ$$

When an unpolarised light is incident on a medium of reflective index 1.73 at Brewster's angle, then reflected light is completely polarised.

41. (d) :



Let potential at junction CD be x

$$\frac{x-50}{1} + \frac{x-50}{3} + \frac{x-0}{2} + \frac{x-0}{4} = 0$$

$$12x - 600 + 4x - 200 + 6x + 3x = 0$$

$$25x - 800 = 0$$

$$\text{or } 25x = 800 \Rightarrow x = 32 \text{ V}$$

Current through 1Ω resistance

$$I_1 = \frac{50-32}{1} = 18 \text{ A}$$

Current through 2Ω resistance

$$I_2 = \frac{32-0}{2} = 16 \text{ A}$$

$\therefore$ Current through branch CD is

$$I_{CD} = I_1 - I_2$$

$$\text{or } I_{CD} = 18 - 16 = 2 \text{ A}$$

42. (c) : Capacitance without dielectric

$$C_0 = \epsilon_0 \frac{A}{d}$$

Capacitance with first dielectric slab,

$$C_1 = \frac{K_1 \epsilon_0 A}{\frac{3}{8}d} = \frac{8K_1 \epsilon_0 A}{3d}$$

Capacitance with second dielectric slab,

$$C_2 = \frac{K_2 \epsilon_0 A}{\frac{d}{2}} = \frac{2K_2 \epsilon_0 A}{d}$$

Capacitance with air,

$$C_3 = \frac{\epsilon_0 A}{\left(d - \frac{3d}{8} - \frac{d}{2}\right)} = \frac{\epsilon_0 A \times 8}{d}$$

$$\text{Equivalent capacitance, } \frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$\text{or } \frac{1}{C_{eq}} = \frac{3d}{8K_1 \epsilon_0 A} + \frac{d}{2K_2 \epsilon_0 A} + \frac{d}{\epsilon_0 A \times 8}$$

$$\text{or } \frac{1}{C_{eq}} = \frac{d}{\epsilon_0 A} \left[\frac{3}{8K_1} + \frac{1}{2K_2} + \frac{1}{8} \right] \quad \dots(ii)$$

$$\text{Given } C_{eq} = 2C_0$$

$$\text{or } \frac{C_0}{C_{eq}} = \frac{1}{2}$$

$$\therefore \frac{\epsilon_0 A}{d} \times \frac{d}{\epsilon_0 A} \left[\frac{3}{8K_1} + \frac{1}{2K_2} + \frac{1}{8} \right] = \frac{1}{2}$$

$$\Rightarrow \frac{3}{8 \times 1.25K_2} + \frac{1}{2K_2} = \frac{1}{2} - \frac{1}{8}$$

$$\Rightarrow \frac{3}{10K_2} + \frac{1}{2K_2} = \frac{3}{8}$$

$$\Rightarrow \frac{3+5}{10K_2} = \frac{3}{8}$$

$$\Rightarrow 64 = 30K_2$$

$$\Rightarrow K_2 = \frac{64}{30} = 2.13 = \frac{K_1}{1.25} \Rightarrow K_1 = 2.66$$

43. (a) : 10 VSD = 9 MSD

$$\therefore 1 \text{ VSD} = \frac{9}{10} \times \text{MSD}$$

$$\text{or } 1 \text{ VSD} = \frac{9}{10} \times 0.1 = 0.09 \text{ cm}$$

$$\text{LC} = 1 \text{ MSD} - 1 \text{ VSD}$$

$$= 0.1 \text{ cm} - 0.09 \text{ cm} = 0.01 \text{ cm}$$

$$\text{Observed reading} = \text{MSR} + (\text{VSD} \times \text{LC})$$

$$= 5 + (8 \times 0.01) = 5.08 \text{ cm}$$

$$\text{Corrected reading} = \text{Observed reading} - \text{zero error}$$

$$= 5.08 - 0.1 = 4.98 \text{ cm}$$

44. (c) : Magnetic moment, $M = IA$

$$\therefore \frac{M_1}{M_2} = \frac{I \times \pi r_1^2}{I \times \pi r_2^2} = \frac{r_1^2}{r_2^2} \quad \text{or} \quad \frac{M_1}{M_2} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

45. (d) : $P_A = P_B = P_O$

$$P_C = P_O - \rho gy$$

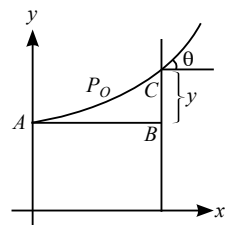
$$P_C = P_O - \frac{S}{R}$$

$$\text{So, } \rho gy = \frac{S}{R}$$

...(i)

$$\text{Now, } R = \frac{\left\{1 + \left(\frac{dy}{dx}\right)^2\right\}^{3/2}}{\frac{d^2y}{dx^2}} \quad (\text{radius of curvature})$$

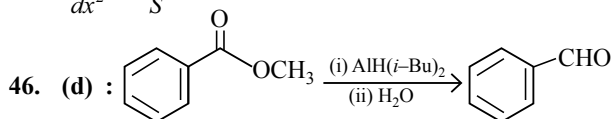
$$\text{As } \frac{dy}{dx} \text{ is small, so } R = \frac{1}{\frac{d^2y}{dx^2}}$$



Putting in eqn. (i),

$$\rho g y = S \times \frac{d^2 y}{dx^2}$$

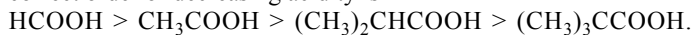
$$\frac{d^2 y}{dx^2} = \frac{\rho g y}{S}$$



Esters are reduced to aldehydes with DIBAL-H.

47. (a) : Acidic strength $\propto -I$ effect $\propto \frac{1}{+I}$ effect. Electron

donating groups decrease the stability of carboxylate ion. Thus, the correct order of decreasing acidity is



48. (b) : $2\text{CuO} + \text{C} \xrightarrow{\Delta} 2\text{Cu} + \text{CO}_2$

The given reaction is used to detect the presence of carbon in organic compounds. Lassaigne's test is used to detect the presence of nitrogen, sulphur, halogens and phosphorus in organic compounds.

49. (c) : Rate constant, $k = 0.03 \text{ s}^{-1}$

For 1st order reaction, $k = \frac{2.303}{t} \log \frac{a}{a-x}$

$$\text{So, } t = \frac{2.303}{0.03} \log \frac{7.2}{0.9} = \frac{2.303}{0.03} \log 8 = \frac{2.303}{0.03} \times 3 \log 2$$

$$= \frac{2.303}{0.03} \times 3 \times 0.301 [\text{Given, } \log 2 = 0.301]$$

$$= 69.32 \text{ s}$$

50. (d) : Diatomic molecule with bond order zero does not exist. With increase in bond order, bond length decreases. Thus, both statement-I and statement-II are false.

51. (c, d) : Conductance of complex compounds in solution depends on number of ions produced. As in both compounds (c) and (d), no ions will be produced, so they will be having the minimum conductance in solution.

52. (a) : Elevation in boiling point, $\Delta T_b = m \times i \times K_b$
or $\Delta T_b \propto m \times i$ (As, K_b is constant.)
or $\Delta T_b \propto \text{concentration} \times \text{no. of ions}.$

For 0.01 M Na_2SO_4 , $\Delta T_b = 0.01 \times 3 \times K_b = 0.03 K_b$

For 0.015 M $\text{C}_6\text{H}_{12}\text{O}_6$, $\Delta T_b = 0.015 \times 1 \times K_b = 0.015 K_b$

For 0.01 M urea, $\Delta T_b = 0.01 \times 1 \times K_b = 0.01 K_b$

For 0.01 M KNO_3 , $\Delta T_b = 0.01 \times 2 \times K_b = 0.02 K_b$

Thus, 0.01 M Na_2SO_4 solution will exhibit highest boiling point.

53. (c)

54. (c) : KO_2 ionises as K^+ and O_2^- ions as it is a superoxide. Thus, oxidation state of K is +1.

H_2O_2 ionises as H^+ and O_2^{2-} as it is a peroxide.

Oxidation state of each O in O_2^{2-} is -1.

In H_2SO_4 , let the oxidation state of S be x ; hence

$$2(+1) + x + 4(-2) = 0$$

$$2 + x - 8 = 0$$

$$x = +6$$

55. (a) : A. Haber's process – I Fe catalyst

B. Wacker oxidation – II PdCl_2

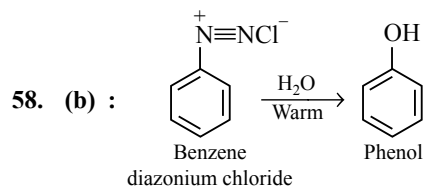
C. Wilkinson's catalyst – III $[(\text{PPh}_3)_3 \text{RhCl}]$

D. Ziegler catalyst – IV TiCl_4 with $\text{Al}(\text{CH}_3)_3$

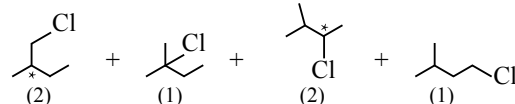
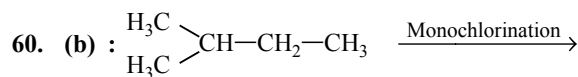
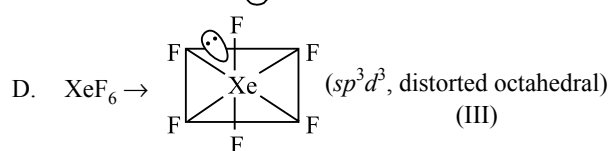
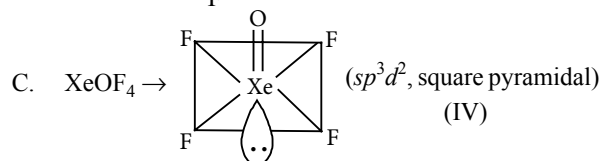
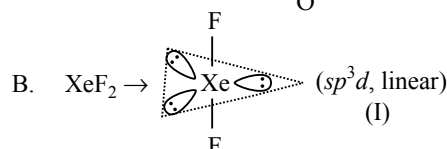
56. (a) : All elements of group-15 can form oxides of E_2O_3 and E_2O_5 type. Thus, antimony can form antimony pentoxide (Sb_2O_5).

57. (a) : Ferromagnetism is an extreme form of paramagnetism hence, statement-I is true.

Electronic configuration of Cr^{2+} : $[\text{Ar}]3d^4$ thus, number of unpaired electrons is 4. Electronic configuration of Nd^{3+} : $[\text{Xe}]4f^3$ thus, number of unpaired electrons is 3. Hence, statement-II is false.



59. (c) : A. $\text{XeO}_3 \rightarrow \text{O}=\text{Xe}=\text{O}$ (sp^3 , pyramidal) (II)



Thus, total six isomers are possible.

61. (d) : A. Ga has melting point of 303 K whereas Cs has melting point of 302 K.

B. On Pauling scale, the electronegativity values of N and Cl are same i.e., 3.0.

C. All the given species are isoelectronic as all have same number of electrons (i.e., 18).

D. The correct order of first ionisation enthalpies is

$$\text{Si} > \text{Mg} > \text{Al} > \text{Na}$$

$$I.E_1(\text{in kJ mol}^{-1}) \quad 786 \quad 737 \quad 577 \quad 496$$

E. As atomic radius increases on going down the group hence the given statement is correct.

62. (c) : $\Delta H_f(\text{BaSO}_4) = \Delta H_f(\text{Ba}^{2+}) + \Delta H_f(\text{SO}_4^{2-}) + \Delta H_{\text{crys.}}(\text{BaSO}_4)$

$$\Delta H_f(\text{Ba}^{2+}) = \Delta H_f(\text{BaSO}_4) - \Delta H_f(\text{SO}_4^{2-}) - \Delta H_{\text{crys.}}(\text{BaSO}_4)$$

$$\Delta H_f(\text{Ba}^{2+}) = -349 + 216 + 4.5 \text{ kcal/mol} = -128.5 \text{ kcal/mol}$$

73. (b) : For a first order reaction,

$$t_{1/2} = \frac{0.693}{k}$$

$$\text{or, } k = \frac{0.693}{t_{1/2}} = \frac{0.693}{1} = 0.693 \text{ min}^{-1} \quad (\because t_{1/2} = 1 \text{ minute})$$

$$\text{For 99.9\% completion of reaction, } k = \frac{2.303}{t} \log \frac{a}{a-x}$$

$$\text{or, } t = \frac{2.303}{k} \log \frac{a}{a-x} = \frac{2.303}{0.693} \log \frac{100}{100-99.9}$$

$$\text{or, } t = \frac{2.303}{0.693} \times 3 = 9.97 \text{ min} \approx 10 \text{ min}$$

74. (c) : $E_n = -2.18 \times 10^{-18} \frac{Z^2}{n^2} \text{ J}$

For first Bohr orbit of He^+ ,

$$E_{(\text{He}^+)} = -2.18 \times 10^{-18} \times \frac{4}{1} = -8.72 \times 10^{-18} \text{ J}$$

For first Bohr orbit of Li^{2+} ,

$$E_{(\text{Li}^{2+})} = -2.18 \times 10^{-18} \times \frac{9}{1} = -19.62 \times 10^{-18} \text{ J}$$

Bohr's radius, $r_n = 52.9 \times \frac{n^2}{Z} \text{ pm}$

For first Bohr orbit of He^+

$$r_{(\text{He}^+)} = 52.9 \times \frac{1}{2} = 26.45 \text{ pm}$$

For first Bohr orbit of Li^{2+} ,

$$r_{(\text{Li}^{2+})} = 52.9 \times \frac{1}{3} = 17.63 \text{ pm}$$

75. (c)

76. (b) : Dalton's atomic theory could not explain the law of gaseous volume.

77. (c) : A. $\text{H}_2\text{O} > \text{NH}_3 > \text{CHCl}_3$: Order of dipole moment

B. $\text{XeF}_2 > \text{XeF}_4 > \text{XeO}_3$: Decreasing number of lone pairs of electrons on central atom

C. $\text{N}-\text{O} > \text{C}-\text{H} > \text{O}-\text{H}$: Order of bond length

D. $\text{N}_2 > \text{O}_2 > \text{H}_2$: Order of bond enthalpy

78. (d) : **Vitamin** **Deficiency disease**

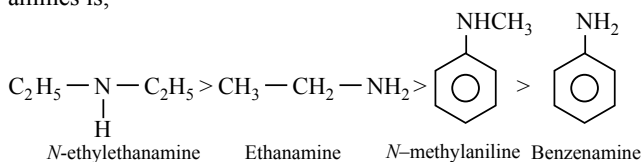
Vitamin B₁₂ Pernicious anaemia

Vitamin D Rickets

Vitamin B₂ Cheilosis

Vitamin B₆ Convulsions

79. (a) : The correct order of decreasing basic strength of given amines is,



Electron releasing groups increase the basic strength of amines.

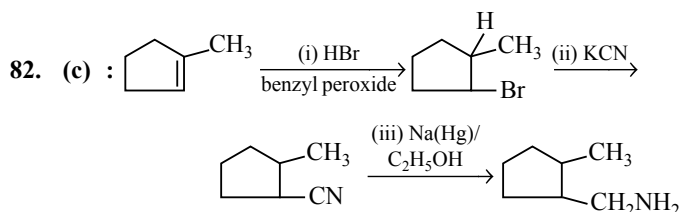
80. (d) : Stronger is the strength of the ligand, greater is the splitting and higher is the energy absorbed by complex.

$$\text{As, } E = \frac{hc}{\lambda} \text{ or } E \propto \frac{1}{\lambda}$$

So, wavelength of light absorbed is inversely proportional to strength of the ligand.

81. (c) : For decolourisation of bromine water, the molecule must be unsaturated hence cyclohexane cannot decolourise bromine water.

Phenol and aniline decolourise bromine water to give white ppt. of 2, 4, 6-tribromoderivatives.



83. (c) :

A. Chloroform + Aniline – Simple distillation (IV)

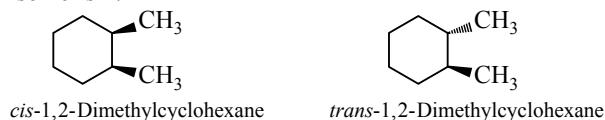
B. Crude oil in petroleum – Fractional distillation (III) industry

C. Glycerol from spent-lye – Distillation under reduced pressure (I)

D. Aniline-water – Steam distillation (II)

84. (d) : The *p*-block elements together with *s*-block elements are known as representative elements or main group elements.

85. (b) : 1,2-Dimethylcyclohexane will show geometrical isomerism.



86. (b) : $K = K_{a_1} \times K_{a_2} \times K_{a_3}$

$$\text{Hence, } \log K = \log K_{a_1} + \log K_{a_2} + \log K_{a_3}$$

It is difficult to remove a proton from negatively charged ion hence, H_3PO_4 is stronger acid than H_2PO_4^- and HPO_4^{2-} .

Higher order ionisation constants (K_{a_2} , K_{a_3}) are smaller than the lower order ionisation constant (K_{a_1}). Hence $K_{a_1} > K_{a_2} > K_{a_3}$.

87. (d) : Co^{2+} : Group – IV (A–III)

Mg^{2+} : Group – VI (B–IV)

Pb^{2+} : Group – I (C–I)

Al^{3+} : Group – III (D–II)

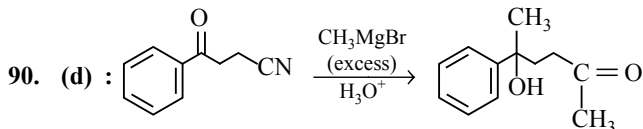
88. (b) : $\text{N}_{2(g)} + \text{O}_{2(g)} \rightleftharpoons 2\text{NO}_{(g)}; \Delta H = +180.7 \text{ kJ mol}^{-1}$

As the given reaction is endothermic hence, it will go to product side on increasing temperature. Also, on adding reactants, reaction goes towards product side.

Hence, conditions given in A, C and D increases the production of $\text{NO}_{(g)}$.

89. (c) : Benzenediazonium chloride is prepared by the reaction of aniline with nitrous acid at 273–278 K. It decomposes easily in the dry state. So, statement-I is correct.

Iodine is not easily introduced into the benzene ring directly thus, iodobenzene is prepared by treatment of benzene diazonium salt with KI. Hence, statement-II is also correct.



91. (a) : The insertion of alien piece of DNA at *EcoRI* site in the given plasmid, will lead to the insertional inactivation. In this, an alien piece of DNA is inserted within the coding sequence of an enzyme, β -galactosidase. This results into inactivation of the gene for synthesis of this enzyme and the colonies do not produce any colour. These colonies are marked as recombinant colonies. If the plasmid in the bacterium does not have an insert, *i.e.*, is non-recombinant, the presence of chromogenic substrate gives blue coloured colonies.

92. (a) : Enzymes could be simple or conjugated (holoenzyme). Conjugated enzymes are formed of two parts- a protein part called apoenzyme and a non-protein part named co-factor. Co-factors are bound to the enzyme to make it catalytically active. There are three types of cofactors: prosthetic groups, co-enzymes and metal ions.

93. (a) : The rate of production of organic matter during photosynthesis in an ecosystem is called gross primary productivity (GPP). A considerable amount of GPP is utilised by autotrophs in respiration. Gross primary productivity minus respiration losses (R) is the net primary productivity (NPP).

$$\text{GPP} - \text{R} = \text{NPP}$$

94. (a) : There is a characteristic distribution of the cells within an unfertilised angiosperm embryo sac. Three cells are grouped together at the micropylar end and constitute the egg apparatus. The egg apparatus, in turn, consists of two synergids and one egg cell. The synergids have special cellular thickenings at the micropylar tip called filiform apparatus, which play an important role in guiding the pollen tubes into the synergid. Three cells are at the chalazal end and are called the antipodals. The large central cell has two polar nuclei. Thus, a typical unfertilised angiosperm embryo sac, at maturity, though 8-nucleate is 7-celled.

95. (d) : Malignant tumors are a mass of proliferating cells called neoplastic or tumor cells. These cells grow very rapidly, invading and damaging the surrounding normal tissues. These cells do not remain confined to original location. Cells sloughed from such tumors reach distant sites through blood and wherever they get lodged in the body, they start a new tumor there. This property called metastasis is the most feared property of malignant tumors. Benign tumors normally remain confined to their original location and do not spread to other parts of the body.

96. (d) : The gymnosperms are plants in which the ovules are not enclosed by any ovary wall and remain exposed, both before and after fertilisation. The seeds that develop post-fertilisation, are not covered, *i.e.*, are naked. Angiosperms bear flowers for sexual reproduction. Seeds are enclosed in fruits in angiosperms.

97. (c) : Progesterone – Corpus luteum

Relaxin – Ovary

Melanocyte stimulating hormone – Pars intermedia

Catecholamines – Adrenal medulla

98. (a) : In human genome, chromosome 1 has the highest number of genes, *i.e.*, 2968 and the Y has the fewest, *i.e.*, 231.

99. (b) : Carboxylation is the fixation of CO_2 into a stable organic intermediate. It is the most crucial step of Calvin cycle where CO_2 is utilised for the carboxylation of RuBP. This reaction is catalysed by enzyme RuBisCO which results in the formation of two molecules of 3-phosphoglyceric acid or 3-PGA, which is the first stable product of photosynthesis. RuBisCO has high affinity for CO_2 when the ratio of CO_2 to O_2 is nearly equal. At mild temperature, RuBisCO has high affinity for CO_2 that is about 80 times higher than its affinity for oxygen. In addition, RuBisCO has higher affinity for O_2 when temperature increases. The phenomenon of breaking up of water into hydrogen and oxygen in the illuminated chloroplasts is called photolysis or photocatalytic splitting of water. Water splitting complex is associated with the PS II, which itself is physically located on the inner surface of thylakoid membrane. There is no role of RuBisCO in photolysis of water.

100. (d) : The first menstruation begins at puberty and is called menarche. Menstrual cycle ceases around 50 years of age in human females, and is called menopause.

101. (c) : In 1983, Eli Lilly an American company, first prepared two DNA sequences corresponding to A and B chains of human insulin and introduced them in plasmids of *Escherichia coli* (enteric bacterium) to produce insulin chains. Chains A and B were produced separately, extracted and combined by creating disulphide bonds to form human insulin (humulin).

102. (c) : Phylum Chordata possesses three sub-phylum (Urochordata, Cephalochordata, Vertebrata) which possess notochord at some stage of their life. The members of subphylum Vertebrata possess notochord during the embryonic period. The notochord is replaced by a cartilaginous or bony vertebral column in the adult. Thus, all vertebrates are chordates but all chordates are not vertebrates.

103. (c) : During metaphase, spindle fibres attach to kinetochores of chromosomes. In anaphase, the two chromatids separate as the spindle fibres attached to the kinetochores shorten and pull the chromosomes to the poles.

104. (a)

105. (a) : Adenosine – Nucleoside

Adenylic acid – Nucleotide

Adenine – Nitrogen base

Alanine – Amino acid

106. (b) : Special venous connection between liver and intestine as well as the kidney and lower parts of the body are present in frogs. The former is called hepatic portal system and the latter is called renal portal system.

107. (d) : Eukaryotic transcription occurs inside nucleus. The mRNA is processed from the primary RNA transcript, a process called maturation. Initially at the 5' end, a cap (consisting of 7-methyl guanosine or 7 mG) and a tail of poly A contains adenine residue (200-300) at the 3' end are added to hnRNA in a template independent manner. The primary mRNAs are made up of two types of segments; non-coding introns and the coding exons. The introns are removed by a process called RNA splicing. The enzyme ligase joins two adjacent exons to produce mature mRNA.

108. (d) : In PCR, a specific segment of DNA amplifies exponentially with each cycle. The basic equation for PCR amplification is 2^n , where n is the number of cycle.

109. (d) : In water and wind pollinated plants, flowers are neither colourful nor produce nectar for pollination. Pollen grains coming in contact with the stigma is a chance factor in both wind and water pollination. To compensate for these uncertainties and associated loss of pollen grains, the flowers produce enormous amount of pollen.

110. (c) : Epiphytes which grow on the branches of mango, cite an example of commensalism. Commensalism is a type of positive interaction in which, two organisms (of different species) live together, and one gets the benefit from the association while the other is neither benefitted nor harmed.

111. (b) : After completion of 12 weeks (first trimester) of gestation period in human beings, most of the major organ systems are formed, for example, the limbs and external genital organs are well developed.

112. (b) : C represents the foam breaker which is used to control foam formation in bio-reactor. Whereas A, B and D represent the flat bladed impeller, motor and sterile air respectively.

113. (a) : The respiration in frogs takes place by different methods in both water and on land. In water, skin acts as aquatic respiratory organ (cutaneous respiration). Dissolved oxygen in the water is exchanged through the skin by diffusion. On land, the buccal cavity, skin and lungs act as the respiratory organs.

114. (d) : The adrenal medulla secretes two hormones adrenaline and noradrenaline, commonly called catecholamines. During emergency, these hormones causes increased activity of heart, dilation of pupil, piloerection (raising of hair) and stimulate breakdown of glycogen resulting hyperglycemic condition.

115. (c) : Absciscic acid is a plant growth inhibitor. Apical dominance is the phenomenon by which presence of apical bud does not allow the nearby lateral buds to grow. Apical bud inhibits the growth of lateral buds by releasing auxins.

116. (d) : Neurohypophysis or pars nervosa also known as posterior pituitary, stores and releases two hormones called oxytocin and vasopressin/ anti-diuretic hormone (ADH). These hormones are actually synthesised by the neurosecretory cells in the hypothalamus and are transported axonally to neurohypophysis.

117. (a) : The yeast, which are used in the brewing industry for the preparation of alcoholic drinks are collectively called brewer's yeast. Different kinds of alcoholic beverages are produced by using different brewer's yeast (*Saccharomyces cerevisiae*) and the fermenting medium. Beer and wine are non-distilled beverages whereas whisky, rum, gin, brandy, vodka and fenny are distilled beverages, so are called 'hard liquors' (higher percentage of alcohol).

118. (d)

119. (c) : Head – Genetic material

Middle piece – Energy

Acrosome - Enzymes

Tail – Sperm motility

120. (a) : When a flower can be divided into two similar halves only in one particular vertical plane, it is zygomorphic, e.g., pea, gulmohar, bean, *Cassia*, etc. *Petunia*, *Datura* and chilli are actinomorphic flowers, i.e., they can be divided into two equal halves in any radial plane passing through the centre.

121. (d) : Cyanobacteria are autotrophic microbes widely distributed in aquatic and terrestrial environments many of which can fix atmospheric nitrogen, e.g., *Anabaena*, *Nostoc*, *Oscillatoria*, etc. Some bacteria can fix atmospheric nitrogen while free-living in the soil (examples, *Azospirillum* and *Azotobacter*), thus enriching the nitrogen content of the soil.

122. (a) : In *ex-situ* conservation approach, threatened animals and plants are taken out from their natural habitat and placed in special setting where they can be protected and given special care. Zoological parks and botanical gardens serve this purpose. Protected areas, wildlife sanctuaries and National parks are methods of *in-situ* conservation.

123. (d) : Ramdeo Misra is revered as the Father of Ecology in India.

124. (c) : RNA was the first genetic material as suggested that essential life processes (such as metabolism, translation, splicing, etc.), evolved around RNA. RNA used to act as a catalyst (there are some important biochemical reactions in living systems that are catalysed by RNA catalysts and not by protein enzymes). But, RNA being a catalyst was reactive and hence unstable. Therefore, DNA has evolved from RNA with chemical modifications that make it more stable genetic material. DNA being double stranded and having complementary strand further resists changes by evolving a process of repair.

125. (b) : *t*RNA and *r*RNA interact with *m*RNA during translation. *m*RNA carries the genetic code, *t*RNA delivers amino acids to the ribosomes and *r*RNA forms the ribosomes that facilitates protein synthesis.

126. (b)

127. (b) : All living members of Class Cyclostomata are ectoparasites on some fishes.

128. (b) : Streptokinase produced by the bacterium *Streptococcus* and modified by genetic engineering is used as a 'clot buster' for removing clots from the blood vessels of patients who have undergone myocardial infarction leading to heart attack.

129. (d) : The most distinctive feature of echinoderms is the presence of water vascular system which helps in locomotion, capture and transport of food and respiration.

130. (d)

131. (b) : Chemotherapeutic drugs are cytotoxic agents that kill or inhibit the growth of rapidly dividing cancer cells. In the case of leukaemia, blood cell counts are increased.

132. (c) : IVF can pose health risks to the mother, though high fatality is rare. IVF requires highly precised handling by specialised professionals and expensive instrumentation. Therefore, these facilities are presently available in very few centres. In IVF, all embryos would probably not survive. Increased use of IVF may reduce the number of people considering adoption. Legal adoption of orphaned children is one of the best methods for couples looking for parenthood.

133. (c)

134. (d) : Asexual reproduction in liverworts, *i.e.*, bryophyte, takes place by formation of specialised structures called gemmae.

135. (c)

136. (c) : Reductionism is method of understanding complex biological systems by breaking them down into smaller and more manageable components to explain biological phenomena at molecular level. This approach focuses on the physico-chemical concept and techniques.

137. (b) : Lymphoid organs are the organs where origin, maturation and proliferation of lymphocytes occur. The primary lymphoid organs are bone marrow and thymus where immature lymphocytes differentiate into antigen-sensitive lymphocytes. Both bone-marrow and thymus provide micro-environments for the development and maturation of T-lymphocytes. After maturation, the lymphocytes migrate to secondary lymphoid organs like spleen, lymph nodes, tonsils and Peyer's patches of small intestine and appendix. The secondary lymphoid organs provide the sites for interaction of lymphocytes with the antigen, which then proliferate to become effector cells.

138. (d)

139. (a)

140. (d) : Innate immunity is a non-specific type of defence present since birth.

141. (d)

142. (c) : Tapetum is the innermost layer of anther wall which completely surrounds the sporogenous tissue. The cells of tapetum are relatively large and centripetally extended having dense cytoplasm and more than one nucleus. Tapetum provides nourishment to the developing microspores, secretes callase enzyme, sporopollenin coated ubisch bodies and pollen kitt.

143. (c) : The eukaryotic ribosomes are 80S while the prokaryotic ribosomes are 70S. Here 'S' stands for the sedimentation coefficient; it indirectly is a measure of density and size. Both 70S and 80S ribosomes are composed of two subunits. The two subunits of 80S ribosomes are 60S and 40S while that of 70S ribosomes are 50S and 30S.

144. (b) : In peroxidase and catalase, which catalyse the breakdown of hydrogen peroxide to water and oxygen, haem is the prosthetic group and it is a part of the active site of the enzyme.

145. (b) : In a frog, the blood vessel that carries deoxygenated blood from the body to the heart is the vena cava. Specifically, the inferior vena cava brings deoxygenated blood from the lower body, while the superior vena cava carries deoxygenated blood from the upper body.

146. (c)

147. (b) : The selection of recombinants and non-recombinants can be done by insertional inactivation. The inactivation of a gene by insertion of a DNA sequence within it is known as insertional inactivation. In this, a recombinant DNA is inserted within the coding sequence of an enzyme, β -galactosidase. This result into inactivation of the gene for synthesis of this enzyme and the colonies do not produce any colour. These colonies are marked as recombinant colonies.

148. (d) : *Aspergillus niger* and *Trichoderma polysporum* are used to produce citric acid and cyclosporin-A respectively. These two are industrial products synthesised with the help of microbes. Citric acid is produced through the fermentation carried out by *Aspergillus niger* on many carbohydrates (such as cane and beet molasses, etc.). Cyclosporin-A is an important bioactive chemical produced by the fungus *Trichoderma polysporum*. It is an 11-membered cyclic oligopeptide. This chemical is used as an immunosuppressive agent in organ transplantation.

149. (c) : RNAi (RNA interference) takes place in all eukaryotic organisms as a method of cellular defense. This method involves silencing of a specific *m*RNA due to a complementary dsRNA molecule that binds to and prevents translation of the *m*RNA (silencing).

150. (d) : The mitochondrial electron transport chain consists of 5 main complex:

Complex I – NADH dehydrogenase complex

Complex II – Succinate dehydrogenase complex

Complex III – Cytochrome *bc*₁ complex

Complex IV – Cytochrome *c* oxidase

Complex V – ATP synthase

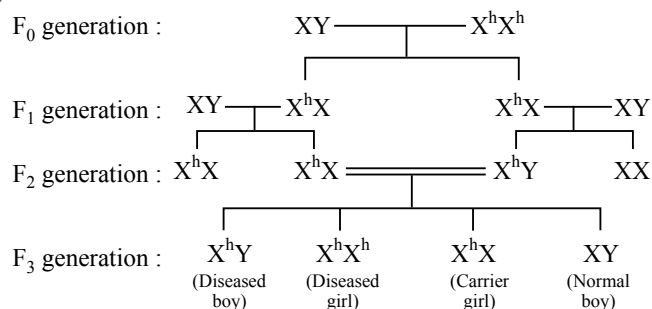
151. (d)

152. (b) : In a floral formula, $\oplus$ stands for actinomorphic nature of the flower and $\underline{G}$ stands for superior ovary.

153. (a) : Flow of energy is unidirectional, *i.e.*, from the sun to producers and then to consumers. This is in accordance with the first law of thermodynamics. According to second law of thermodynamics, every activity involving energy transformation is accompanied by dissipation of energy. In other words, the loss of biologically useful energy as heat with every energy transfer in an ecosystem is a consequence of the second law of thermodynamics. Every time energy is transferred from one place to another or transformed from one kind to another, some of it is converted into heat.

154. (b) : Productivity refers to the rate of biomass production per unit area over a time period by plants, *i.e.*, the rate at which sunlight is accumulated by producers for the synthesis of energy rich organic compounds and is expressed in terms of ($\text{Kcal m}^{-2} \text{ yr}^{-1}$ (measured in terms of energy) or $\text{gm}^{-2} \text{ yr}^{-1}$ (measured in terms of weight)).

155. (c) : The probability for the birth of a child having no disease and being a carrier (has the disease mutation in one allele of the gene) in F_3 generation would be $1/4$. This can be illustrated by the given cross:



156. (b) : In seeds of cereals, the food storage tissue endosperm is rich in starch and a protein rich aleurone layer lies on the outside of endosperm.

157. (d) : Chlorophyll *a* – Blue-green

Chlorophyll *b* – Yellow-green

Xanthophylls – Yellow

Carotenoids – Yellow to yellow-orange

158. (c) : It was George Gamow, a physicist, who argued that since there are only 4 bases and if they have to code for 20 amino acids, the code should constitute a combination of bases. He suggested that in order to code for all the 20 amino acids, the code should be made up of three nucleotides.

159. (c) : Histones are water-soluble proteins rich in arginine and lysine that help in DNA packaging in chromosomes. Prokaryotic cells lack histone proteins.

160. (c) : DNA mutase is an enzyme that induces mutation in DNA, while DNA recombinase is involved in recombination. These two enzymes are not typically used in standard gene cloning procedures.

161. (c) : Mesosome is a special membranous structure found in prokaryotes which is formed by the extensions of plasma membrane into the cell. These extensions help in cell wall formation, DNA replication, respiration, secretion processes, etc.

162. (a) : ρ (rho) is a terminator factor that terminates the transcription.

163. (b) : Copulatory pad is present on the first digit of the forelimbs of male frog.

164. (d)

165. (a) : Antigen-binding site is located at the N-terminal region of antibody molecules.

166. (c) : Scutellum – Cotyledon of monocot seed

Non-albuminous seed – Groundnut

Epiblast – Rudimentary cotyledon

Perisperm – Persistent nucellus

167. (c) : In monocot stem, hypodermis is sclerenchymatous.

168. (d) : Since the twins born are of different sexes, hence they are fraternal twins. This is because fraternal twins occur when two separate eggs are fertilised by two different sperms, resulting in different sexes of the newly born twins.

169. (c) : Sweet potato and potato are analogous structures. Analogous structures are the result of convergent evolution.

170. (b) : Cytokinins are phytohormones that promote nutrient mobilisation which helps in the delay of leaf senescence in plants.

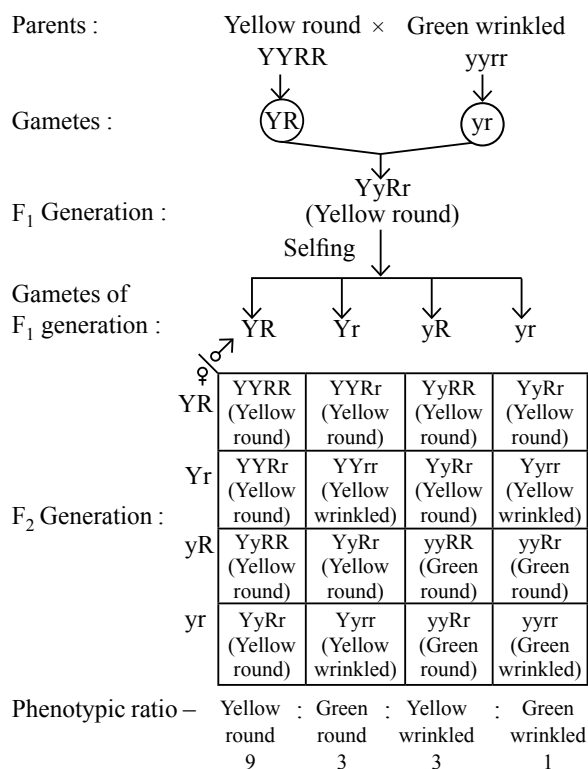
171. (d) : Insulin cannot be given orally to diabetic patients because it is a protein that will be digested in gastro-intestinal tract and will not reach the bloodstream in its active form.

172. (a) : Transferase enzyme catalyses a transfer of a group, G (other than hydrogen) from one substrate (S) to another ($S^{\#}$).

173. (c)

174. (a) : The main plant body of the bryophyte is haploid. It produces gametes, hence is called a gametophyte. The gametophyte gets attached to the substratum by unicellular or multicellular rhizoids. The sex organs in bryophytes are multicellular. The male sex organ is called antheridium. They produce biflagellate antherozoids. The female sex organ called archegonium is flask-shaped and produces a single egg. The antherozoids are released into water where they come in contact with archegonium. An antherozoid fuses with the egg to produce the zygote. Zygotes do not undergo reduction division immediately. They produce a multicellular body called a sporophyte. The sporophyte is not free-living but attached to the photosynthetic gametophyte and derives nourishment from it. Some cells of the sporophyte undergo reduction division (meiosis) to produce haploid spores. These spores germinate to produce gametophyte.

175. (a) : The phenotypic ratio of F_2 generation would be $9 : 3 : 3 : 1$. This can be illustrated as follows:



176. (d) : 'A' represents Kingdom Fungi, 'B' Kingdom Animalia,

'C' Kingdom Monera, 'D' Kingdom Plantae and 'E' Kingdom Protista.

177. (b) : Centromere – Cell division

Cilium – Cell movement

Cristae – Mitochondrion

Cell membrane – Phospholipid bilayer

178. (d) : Verhulst-Pearl logistic growth is described by the following equation:

$$dN/dt = rN \left(\frac{K - N}{K} \right)$$

Where N = Population density at time t

r = Intrinsic rate of natural increase

K = Carrying capacity

179. (b) : Emphysema is a chronic disorder in which alveolar walls are damaged due to which respiratory surface is decreased. Angina pectoris is acute chest pain that occurs when enough oxygen is not reaching the heart muscles. Glomerulonephritis is the inflammation of glomeruli of kidney. Tetany is rapid spasm in muscles due to low Ca^{++} in body fluid.

180. (d)



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